

Diagrammatic Monte Carlo

MSci Project Report

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Abstract

Diagrammatic Monte Carlo (DiagMC) is a simulation method that solves quantities of interest in terms of diagrammatic expansions through stochastic sampling. In this paper, we calculate the scattering length for two different scattering systems using the bold line diagrammatic Monte Carlo method (BLDMC) which is a variation of DiagMC. The first scattering system is governed by the square well potential while the second is governed by the Yukawa potential. The results from the BLDMC method were then compared with the scattering length calculated using Numerov's numerical method. These results show that the BLDMC method converges well with the Numerov results for potential values before the first bound state in each system. However, to accurately describe the scattering length past this bound state, it was important to start simulations after the first bound state. The usefulness of DiagMC is highlighted through results showing that for both scattering systems the approximated functions were expressed self-consistently in terms of themselves.

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Chapter 1

Introduction

Diagrammatic expansions appear in important parts of scattering physics, for example interactions between particles in Feynman diagrams [1]. Feynman diagrams can be written in terms of a Dyson series, which is a perturbative expansion where each term can be represented by a sum of Feynman diagrams. Diagrammatic Monte Carlo stochastically samples from each diagram in a configuration space by building a Markov chain based on the Metropolis-Hastings algorithm [2, 3]. This algorithm generates updates that dictate which variables from each diagram are being sampled [4, 5]. A variation of this method, called bold line diagrammatic Monte Carlo (BLDMC), takes the same approach. However, BLDMC attempts to reduce the amount of diagrams calculated explicitly by taking advantage of expansions that can be written in terms of themselves [5]. In this paper we demonstrate the usefulness of DiagMC by calculating the scattering length for two scattering systems.

We start by introducing scattering theory and detailing the two scattering systems that are investigated in this paper, the first governed by the square well potential and the second the Yukawa potential. A numerical approach, Numerov's method, is then explored to calculate parameters for both scattering systems which are then compared to results found through the BLDMC method. Next we discuss Markov chains and the driving stochastic process behind DiagMC. An example of building a Markov chain with the Maxwell-Boltzmann distribution as the target distribution is considered and used to determine the expectation value of the velocity and normalisation constant. This method is then adapted to show how DiagMC can alleviate problems that occur within the Metropolis-Hastings algorithm. Finally, we introduce DiagMC and BLDMC fully, and calculate the scattering length for both systems, comparing answers with solutions found by Numerov's method.

Chapter 2

Scattering Theory

Scattering theory investigates the consequences of “firing” particles incident on a particular area of interest, be that another particle or a potential. By examining the particles after interacting with the object, we can then obtain information about the object itself. In this paper, three-dimensional scattering systems governed by a square well potential (2.14) and Yukawa potential (2.37), were investigated. This chapter explores the general theory behind scattering, the analytical and numerical solutions to the phase shift and scattering length for the square well potential, and lastly the numerical solution to the Yukawa potential system.

2.1 Quantum Scattering in Three Dimensions

Before discussing quantum scattering it is useful to build a classical scattering picture, illustrated in Figure 2.1. In the classical picture, we have a flux of particles incident on a scattering centre along the z -axis, where each particle follows a defined trajectory. The impact parameter b (seen in Figure 2.1), is defined as the perpendicular distance between the initial trajectory and the scattering centre. We define the cross-sectional area denoted $d\sigma$ (seen in Figure 2.1), as the infinitesimally small area associated with particles that are scattered into a solid angle $d\Omega$ (also seen in Figure 2.1). Once the particles are sufficiently close to the scattering centre to experience a potential, they are each deflected by an angle θ . As $b \rightarrow \infty$, the deflection angle vanishes ($\theta \rightarrow 0$), and the particles pass through undeflected. However, for cases where deflection occurs, particles scatter off of the object at an angle θ into a cone of solid angle $d\Omega$ (shown in Figure 2.1).

While the classical picture provides useful intuition about scattering theory, in order to discuss how particles interact microscopically with scattering objects it's necessary to describe these systems using quantum mechanics. Therefore, we consider the time-independent Schrödinger equation for a particle incident on a scattering surface [6]

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(r) \right] \psi(\vec{r}) = E\psi(\vec{r}), \quad (2.1)$$

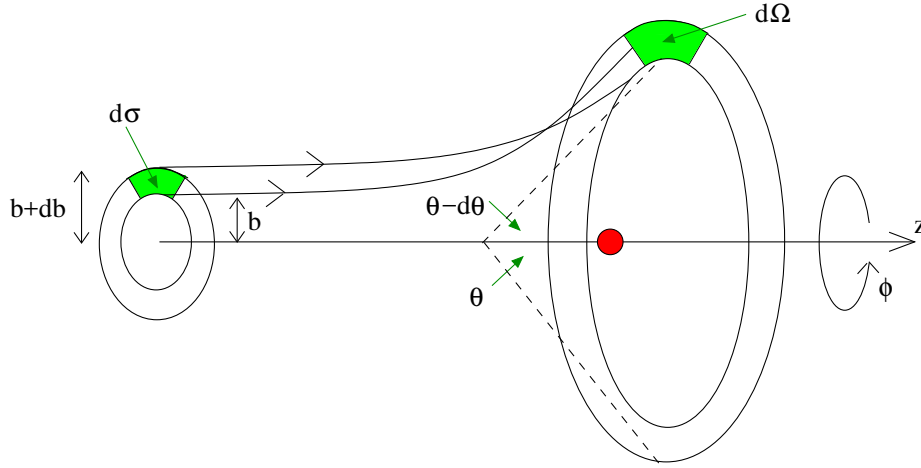


Figure 2.1: Schematic showing particles scattering off of a scattering centre (red ball) and the parameters involved in describing this process [7].

where it is assumed here that the scattering object is a potential and therefore has no mass. However, if the object was another particle, the reduced mass could be used to transform the problem from a complex two-body problem into a single body problem. Once the particle comes into contact with the potential, the goal is to determine how the potential influences the behaviour of the particle. However, close to the potential this is a complicated interaction. Instead, we focus on the wavefunction, post scattering, at large distances from the scattering event origin, such that the potential is negligible. This results in the following asymptotic boundary condition

$$\psi(r, \theta) \sim e^{ikz} + f(\theta) \frac{e^{ikr}}{r} \quad (2.2)$$

where $f(\theta)$ is the scattering amplitude. Equation 2.2 makes the assumption that the potential is spherically symmetrical, removing ϕ dependence on $f(\theta)$. The first term in Equation 2.2 is the incident plane wave moving along the z axis towards the scattering centre while the second term is the scattered wave, which propagates outward as a spherical wave.

This paper covers the square well and Yukawa potentials, both being central potentials (spherically symmetrical). Therefore, it's convenient to express Equation 2.1 in spherical coordinates

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial \psi}{\partial \theta} \right) \right] + V(r) \psi(r, \theta) = E \psi(r, \theta) \quad (2.3)$$

where we have removed ϕ dependence. This also results in the magnetic quantum number m_l being set equal to zero. Equation 2.1 is solved by separation of variables (Derived in Appendix A)

$$\psi(r, \theta) = \sum_l R_l(r) P_l(\cos \theta) \quad (2.4)$$

where $P_l(\cos \theta)$ are the Legendre polynomials, l is angular momentum quantum number, and $R_l(r)$ is the radial wavefunction that satisfies the radial differential equation (Appendix A)

$$\left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{l(l+1)}{r^2} - \frac{2mV(r)}{\hbar^2} + k^2 \right) R_l(r) = 0, \quad \text{where } k = \sqrt{2mE}/\hbar. \quad (2.5)$$

The term $l(l+1)/r^2$ from Equation 2.5 can be interpreted as the centrifugal barrier, which together with $V(r)$ defines the effective potential

$$V_{\tilde{}}(r) = V(r) + \frac{\hbar^2 l(l+1)}{2mr^2}. \quad (2.6)$$

The centrifugal term becomes very large as $r \rightarrow 0$ for $l > 0$, which acts as a repulsive barrier near the origin. As a result, scattering at low energies is dominated by s-waves ($l = 0$). Applying the asymptotic boundary condition to Equation 2.5 produces the following

$$\left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{l(l+1)}{r^2} + k^2 \right) R_l(r) = 0, \quad (2.7)$$

where in the asymptotic region, the potential value is effectively zero $V(r) \approx 0$. Equation 2.7 has the general solution given in the form of spherical Bessel and Neumann functions $j_l(kr)$ and $n_l(kr)$ respectively [6]

$$R_l(r) = B_l(k)j_l(kr) + C_l(k)n_l(kr), \quad (2.8)$$

where $B_l(k)$ and C_l are constants independent of r . Explicitly, the spherical Bessel and Neumann functions are [6]

$$j_l(x) = (-x)^l \left(\frac{1}{x} \frac{d}{dx} \right)^l \frac{\sin(x)}{x} \quad n_l(x) = (-x)^{l+1} \left(\frac{1}{x} \frac{d}{dx} \right)^l \frac{\cos(x)}{x}. \quad (2.9)$$

The spherical Bessel and Neumann functions can also be expressed asymptotically, when $x \rightarrow \infty$

$$j_l(x) \sim \frac{\sin(x - \frac{l\pi}{2})}{x} \quad n_l(x) \sim -\frac{\cos(x - \frac{l\pi}{2})}{x}, \quad (2.10)$$

we substitute these asymptotic versions into Equation 2.8, resulting in

$$R_l(r) \sim \frac{1}{kr} \left[B_l(k) \sin \left(kr - \frac{l\pi}{2} \right) - C_l(k) \cos \left(kr - \frac{l\pi}{2} \right) \right]. \quad (2.11)$$

We can simplify further by rewriting Equation 2.11 as (derived in Appendix B)

$$R_l(r) \sim \frac{A_l(k)}{kr} \sin \left(kr - \frac{l\pi}{2} + \delta_l(k) \right) \quad \text{where } A_l(k) = \sqrt{B_l^2(k) + C_l^2(k)}. \quad (2.12)$$

This puts us in the position to derive useful parameters to describe both scattering systems. For instance, the phase shift (Appendix B)

$$\delta_l(k) = -\arctan \left(\frac{C_l(k)}{B_l(k)} \right). \quad (2.13)$$

The phase shift of the particle's wavefunction is a parameter that quantifies the change in phase induced by the potential. It is best described visually, as seen in Figure 2.2, which illustrates two waves: one that has been deflected by a square well radius potential (black), and one that passes through unaffected (red). The difference between two points (green dots) that are the same on each wave (e.g. crest to crest) describes the phase shift.

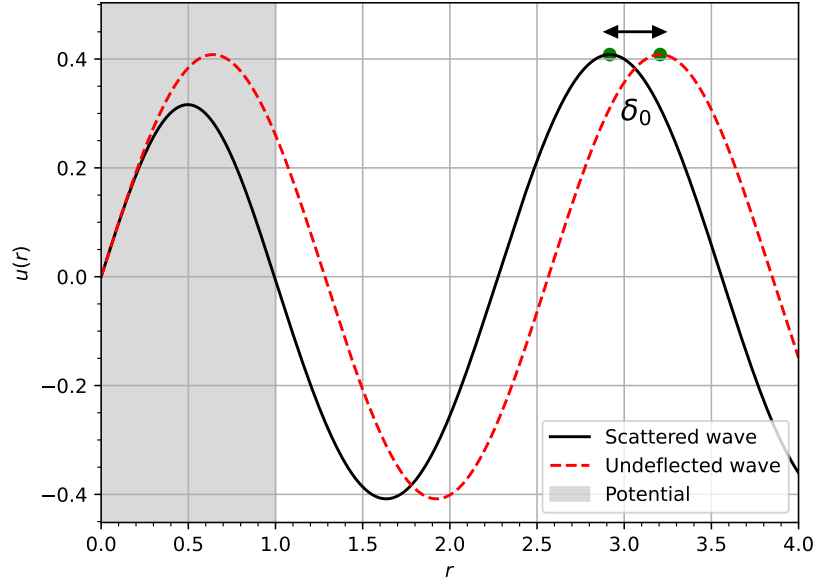


Figure 2.2: This figure displays two s-waves ($l = 0$): one that is affected by the square well potential (black), with a value of 2 Ha (Hartrees) and one that does not experience any potential and passes through unaffected (red). The symbol δ_0 denotes the $l = 0$ phase shift. Both waves were generated by solving Equation 2.29 via Numerov's method, discussed in Section 2.3, in atomic units. The grey region indicates the range of effect that the potential has with the radius of the potential being set equal to one Bohr radius a_0 . Finally, the energy of both waves was set equal to 3 Ha.

Up to this point, we have developed the scattering formalism for a general central potential $V(r)$ that vanishes as $r \rightarrow \infty$, without specifying its explicit form. However, only a limited number of potentials produce fully analytical solutions, one example being the square well potential discussed in Section 2.2. For potentials that don't have analytical solutions, the phase shifts and other parameters must instead be obtained via approximation methods, such as the Born approximation or numerical methods. Rather than continuing with a general undefined potential, we now investigate the square well potential.

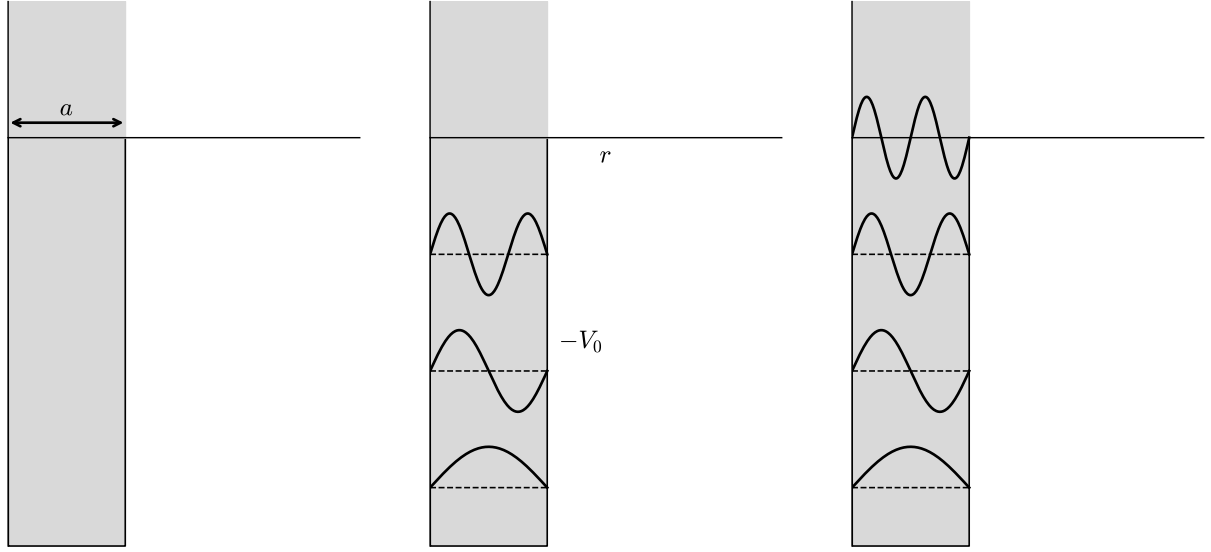


Figure 2.3: This figure is a schematic of the square well potential, where $-V_0$ is the constant value from Equation 2.14, a is the radius of the well, and r is the distance from the origin of the well. The first figure (left) shows the square well with a particular well depth $-V_0$ and radius a . The second figure (middle) shows that the well can hold bound states, where three states are visible. If the well depth is increased it allows for more bound states. Finally, the third figure (right) shows a situation where the well depth is increased such that a bound state is at the threshold of $E = 0$, when this occurs, divergence in parameters such as the scattering length is observed. The grey region shown in each figure displays the range of r values that are affected by the potential.

2.2 Analytical Square Well Potential

The **attractive** square well potential (radius a) is illustrated in Figure 2.3 and is expressed mathematically as

$$V(r) = \begin{cases} -V_0 & \text{if } r < a \\ 0 & \text{if } r > a. \end{cases} \quad (2.14)$$

In this section we look at the analytical solutions for the phase shift δ_l and the scattering length α . Starting from Equation 2.5, we define the radial equation inside the well ($r < a$)

$$\left[\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{l(l+1)}{r^2} + K^2 \right] R_l(r) = 0 \quad (2.15)$$

where $K^2 = k^2 + (2m/\hbar^2)V_0$. Equation 2.15 has the exact same form as Equation 2.7 with k^2 being replaced with K^2 . The following is the solution to Equation 2.15 that describes the interior (indicated with a superscript I) of the square well [6]

$$R_l^I(K, r) = N_l(K) j_l(Kr) \quad (2.16)$$

where $N_l(K)$ is a normalisation constant and $j_l(Kr)$ is the spherical Bessel function (Equation 2.9). Now the solution associated with outside of the well is examined, the exterior region. This corresponds to the potential value (from Equation 2.14) being equal to zero, hence, the exterior radial equation takes the form of Equation 2.7 which has a general solution (Equation 2.8) derived in Section 2.1. Rearranging Equation 2.13, solving for $C_l(k)$ and substituting this into Equation 2.8, the exterior radial wavefunction can be written in the following convenient way

$$R_l^E(k, r) = B_l(k)[j_l(kr) - \tan \delta_l(k)n_l(kr)]. \quad (2.17)$$

To solve for the phase shift we make use of the requirement that at the boundary $r = a$, where the interior and exterior solutions, as well as their derivatives, both equal each other respectively

$$R_l^I(K, a) = R_l^E(k, a) \quad (2.18)$$

$$\left. \frac{dR_l^I(K, r)}{dr} \right|_{r=a} = \left. \frac{dR_l^E(k, r)}{dr} \right|_{r=a}. \quad (2.19)$$

Dividing Equation 2.19 by Equation 2.18 and solving for $\tan \delta_l(k)$, results in

$$\tan \delta_l(k) = \frac{kj'_l(ka)j_l(Ka) - Kj_l(ka)j'_l(Ka)}{kn'_l(ka)j_l(Ka) - Kn_l(ka)j'_l(Ka)} \quad (2.20)$$

$$\text{where} \quad j'_l(x) = \frac{dj_l(x)}{dx} \quad \text{and} \quad n'_l(x) = \frac{dn_l(x)}{dx}$$

therefore, the phase shift for a general l is

$$\delta_l(k) = \arctan \left(\frac{kj'_l(ka)j_l(Ka) - Kj_l(ka)j'_l(Ka)}{kn'_l(ka)j_l(Ka) - Kn_l(ka)j'_l(Ka)} \right). \quad (2.21)$$

Focusing on low-energy interactions only ($l = 0$) allows for the calculation of the scattering length α , which is a parameter that characterises low-energy interactions. The scattering length is of particular interest since it can be utilised to distinguish when 'zero-energy resonance' occurs [6] (discussed in depth in the following analysis). Setting $l = 0$ in Equation 2.20, where

$$j_0(x) = \frac{\sin x}{x} \quad (2.22)$$

$$n_0(x) = -\frac{\cos x}{x} \quad (2.23)$$

$$j'_0(x) = \frac{x \cos(x) - \sin(x)}{x^2} \quad (2.24)$$

$$n'_0(x) = \frac{x \sin(x) + \cos(x)}{x^2} \quad (2.25)$$

we find (derived in Appendix C)

$$\tan \delta_0(k) = \frac{k \tan(Ka) - K \tan(ka)}{K + k \tan(ka) \tan(Ka)}. \quad (2.26)$$

The definition of the scattering length [6] as a mathematical expression is

$$\alpha = -\lim_{k \rightarrow 0} \frac{\tan \delta_0(k)}{k}. \quad (2.27)$$

Using this definition and Equation 2.26 an expression is found for the scattering length that depends solely on the potential V_0 and the radius of the square well a (derived in Appendix C)

$$\alpha = \left(1 - \frac{\tan \gamma}{\gamma}\right)a \quad \text{where} \quad \gamma = a \left(\frac{2mV_0}{\hbar^2}\right)^{1/2}. \quad (2.28)$$

The scattering length for the square well potential is shown in Figure 2.4 where different values of γ correspond to the first three resonance points. These divergences occur due to the presence of a bound state appearing exactly at the threshold $E = 0$. At these points, the scattering length becomes infinite and indicates the formation of a new bound state. This is illustrated in the rightmost graph in Figure 2.3. The first resonance point corresponds to the first bound state, the second resonance corresponds to the second bound state, and so on.

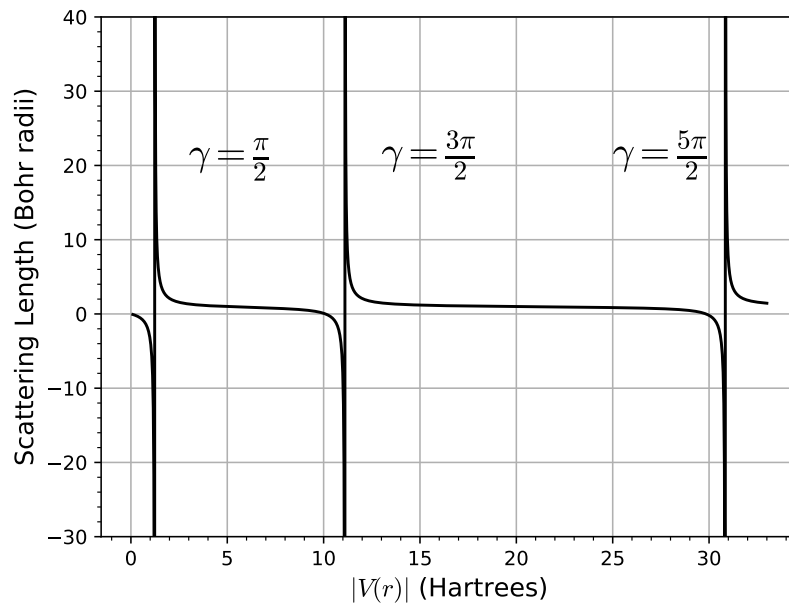


Figure 2.4: Analytical scattering length using Equation 2.28 in atomic units, the figure shows three values of γ that indicate the value of γ at which resonance occurs.

2.3 Numerov's Method

In this section we describe Numerov's numerical method, which is a second order differential equation solver. This method is used to calculate various parameters in both the square well

potential (Equation 2.14) system and Yukawa potential (Equation 2.37) system. We then compare with analytical results shown in Section 2.2 for the square well system.

We begin with finding the reduced radial equation from the radial equation seen in Equation 2.5, this involves defining $u_l(r)/r = R_l(r)$ and substituting it into the radial equation, whereby the first order term is canceled out

$$\frac{d^2 u_l}{dr^2} + \left[k^2 - \frac{l(l+1)}{r^2} - \frac{2mV(r)}{\hbar^2} \right] u_l(r) = 0. \quad (2.29)$$

Equation 2.29 is then solved via Numerov's method (derived in Appendix D)

$$u_{n+1} = \frac{2 \left(1 - \frac{5h^2}{12} g_n \right) u_n - \left(1 + \frac{h^2}{12} g_{n-1} \right) u_{n-1}}{\left(1 + \frac{h^2}{12} g_{n+1} \right)} + \mathcal{O}(h^6). \quad (2.30)$$

$$g_n = \left[k^2 - \frac{l(l+1)}{r_n^2} - \frac{2mV(r_n)}{\hbar^2} \right]$$

where $u_n \equiv u_l(r_n)$ and h is the spacing between our spatial points $r_0, r_1, \dots, r_{N-1}$, i.e. $|r_n - r_{n-1}| = h$. This results in a discretised solution space $u_0, u_1, \dots, u_{N-1}$. The choice of Numerov's method is due to Equation 2.29 possessing zero first order terms. In such cases, this method achieves a truncation error of order $\mathcal{O}(h^6)$ (Appendix D). The first two points (u_0 and u_1) must be initialised for Equation 2.30 to compute u_2 . These are chosen as $u_0 = 0$ and $u_1 = h^{(l+1)}$ (discussed further in Appendix D.1) for both potentials studied. This method is then implemented recursively, given u_{m-1} and u_m the value u_{m+1} is computed. This process is repeated iteratively i.e. the next two points: u_m and u_{m+1} would then be used to generate u_{m+2} and so on. Pseudo code for Numerov's method applied to the reduced radial equation is seen in Appendix D.2.

To calculate the phase shift, we solve for $u_l(r)$ up to and including the asymptotic region, where the potential is approximately zero. We choose two positions r_p and r_q that correspond to the extrema points $u(r_p)$ and $u(r_q)$, as seen in Figure 2.5, where the two red points in both graphs are the chosen extreme points. Both points r_p and r_q , since they are in the asymptotic region, can be approximated as

$$u_l(r_p) \approx Akr_p [\cos(\delta_l)j_l(kr_p) - \sin(\delta_l)n_l(kr_p)] \quad (2.31)$$

$$u_l(r_q) \approx Akr_q [\cos(\delta_l)j_l(kr_q) - \sin(\delta_l)n_l(kr_q)]. \quad (2.32)$$

These take the form from Equation 2.8, where $B_l = A \cos(\delta_l)$ and $C_l = -A \sin(\delta_l)$, and specifically solve for the reduced radial wavefunction¹. From this we can find the phase shift of the

¹The general solution for the reduced radial equation is $u_l(r) = r[B_l(k)j_l(kr) + C_l(k)n_l(kr)]$.

system (derived in Appendix E)

$$\delta_l = \arctan \left(\frac{G j_l(r_p) - j_l(r_q)}{G n_l(r_p) - n_l(r_q)} \right) \quad (2.33)$$

$$\text{where} \quad G = \frac{r_p u_l(r_q)}{r_q u_l(r_p)}. \quad (2.34)$$

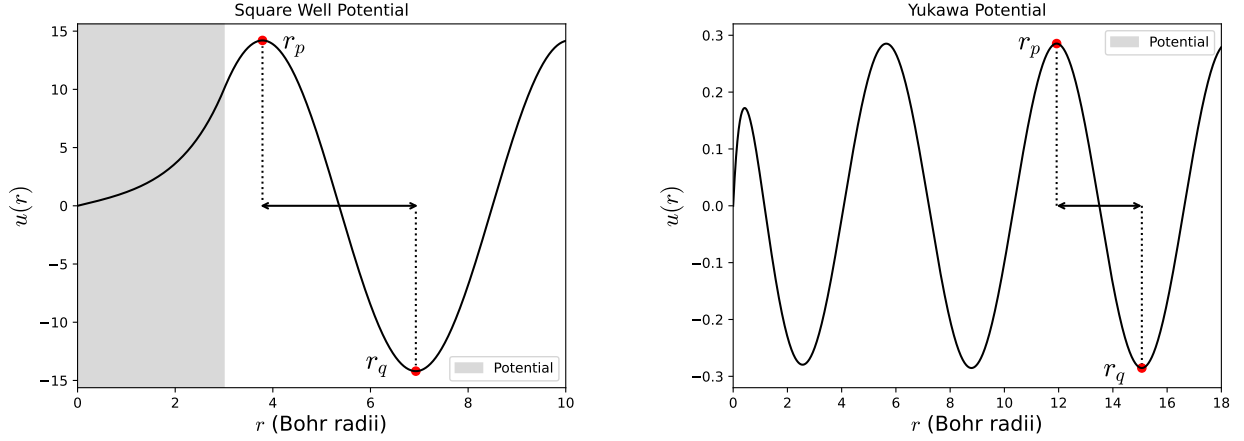


Figure 2.5: Each graph shows a wave being deflected by a potential: square well potential (left), Yukawa potential (right), where the region in grey is the range of effect of the potentials. However, the Yukawa grey region shows the strength as well. Due to the square well potential being a flat value $-V_0$, this isn't shown.

Both graphs feature two red dots, where the waves maximum and minimum points are, the distance of the arrow drawn between the two points is half the wavelength. The parameters used to obtain these graphs were:

Square well: $l = 0$, $E = 0.5$, $a = 3$, $V_0 = 1$. Yukawa: $l = 0$, $E = 0.5$, $g = 2$, $\mu = 1$.

In order to calculate the scattering length numerically, we use the analytical definition of the scattering length (see Equation 2.27), for small enough k values or equivalently E values. The approximation for the scattering length is then

$$\alpha \approx -\frac{\delta_0}{k} \quad (2.35)$$

where the small angle approximation for $\tan(\delta_0)$ has been used.

Having established the application of Numerov's method for computing phase shifts and corresponding scattering lengths for a given general system, we now consider in detail the square well potential, followed by the Yukawa potential.

2.4 Square Well Potential Results

This section details the results obtained from applying Numerov's method to a scattering system defined by the square well potential, where all results are given in atomic units.

Figure 2.6 shows the phase shift calculated for $l = 0, 1, 2, 3$ using Numerov's method (Equation 2.33) compared against analytical results (Equation 2.21). These results show jumps in π for $l = 0, 1$, this is because $\arctan(x)$ is modulo π . The error in the results was calculated using the root mean square (RMS)

$$\text{RMS} = \sqrt{\frac{1}{N_v} \sum_i (\vec{v}_{(num)} - \vec{v}_{(ann)})^2} \quad (2.36)$$

where N_v is the number of elements in vector $\vec{v}_{(num)}$ or $\vec{v}_{(ann)}$. The numerical solutions are represented by the vector $\vec{v}_{(num)}$ and the analytical solutions: $\vec{v}_{(ann)}$. The errors obtained are seen in Table 2.1.

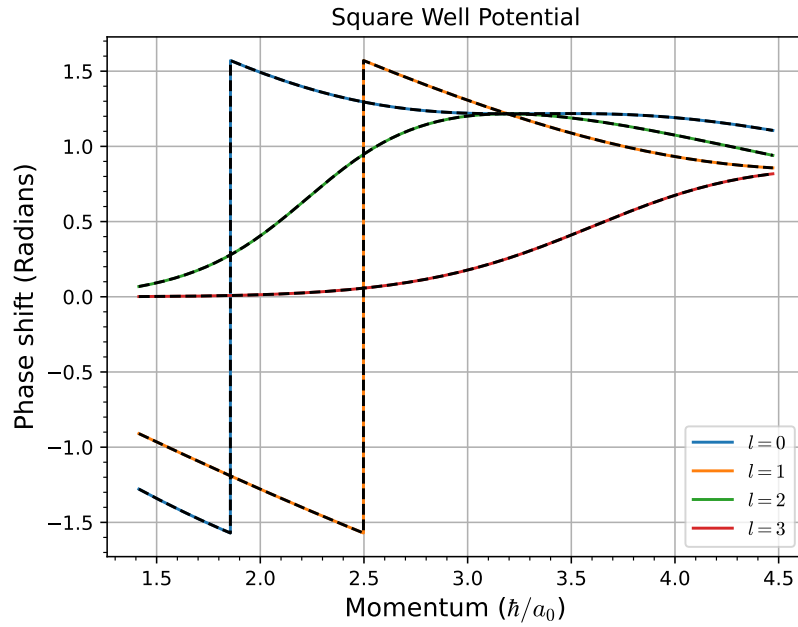


Figure 2.6: The phase shift for the scattering system governed by the square well potential, values for $l = 0, 1, 2, 3$ were calculated using Numerov's method, these are compared with the analytical solution that is displayed as the dashed black lines. The parameters of the system were: $E = 3$, $V_0 = -5$, $a = 1$, $h = 0.00001$

l	RMS error
0	0.000103
1	0.0000346
2	0.000172
3	0.0000136

Table 2.1: RMS (Equation 2.36) error for the phase shifts calculated in Figure 2.6 for different values of l .

The scattering length was subsequently calculated (Equation 2.35) and compared with analytical results (Equation 2.28), shown in Figure 2.7

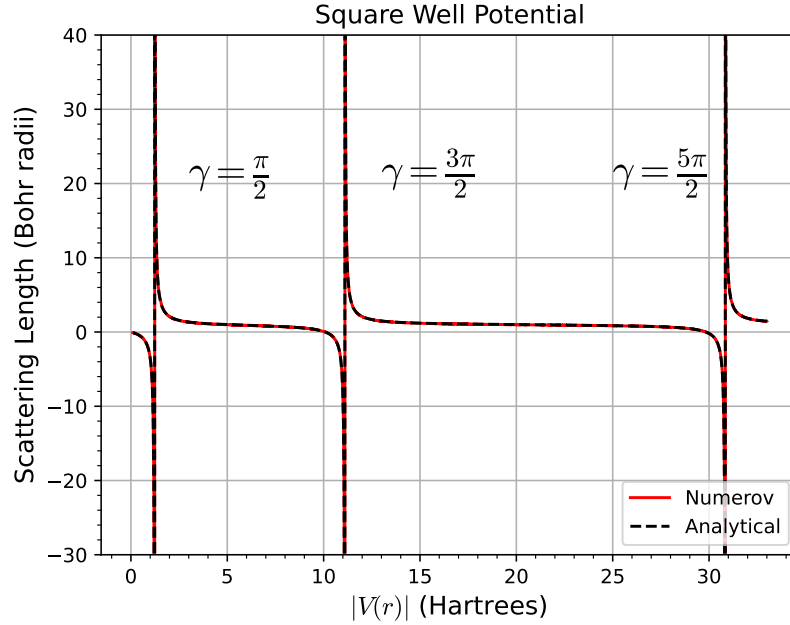


Figure 2.7: This figure shows the comparison between the numerical (Numerov's method discussed in Section 2.3, using Equation 2.35) and analytical (Equation 2.28) calculations of the scattering length for the square well potential. The parameters used were: $l = 0$, $E = 1 \times 10^{-4}$, $h = 0.00001$.

The advantages associated with calculating results for the square well potential were: the points r_p and r_q (discussed in Section 2.3) could be taken immediately after r exceeded the radius (a) of the square well potential (Equation 2.14), meaning the very first appearance of a minimum or maximum value would become r_p . The second advantage is that we had analytical data to compare it to. We will come to see in the next section that the Yukawa potential does not share these advantages.

2.5 Yukawa Potential Results

This section details the results obtained from applying Numerov's method to the scattering system defined by the Yukawa potential, where all results are given in atomic units.

The Yukawa potential, also known as the screened Coulomb potential, is expressed as [6]

$$V_y(r) = g \frac{e^{-\mu r}}{r} \quad (2.37)$$

where g is the coupling constant or coupling strength, μ is a scaling factor, and m is the mass of

the particle. In the analysis below, we have set μ and m equal to unity. The Yukawa potential is also illustrated in Figure 2.8, where darker values of blue correspond to smaller values of g .

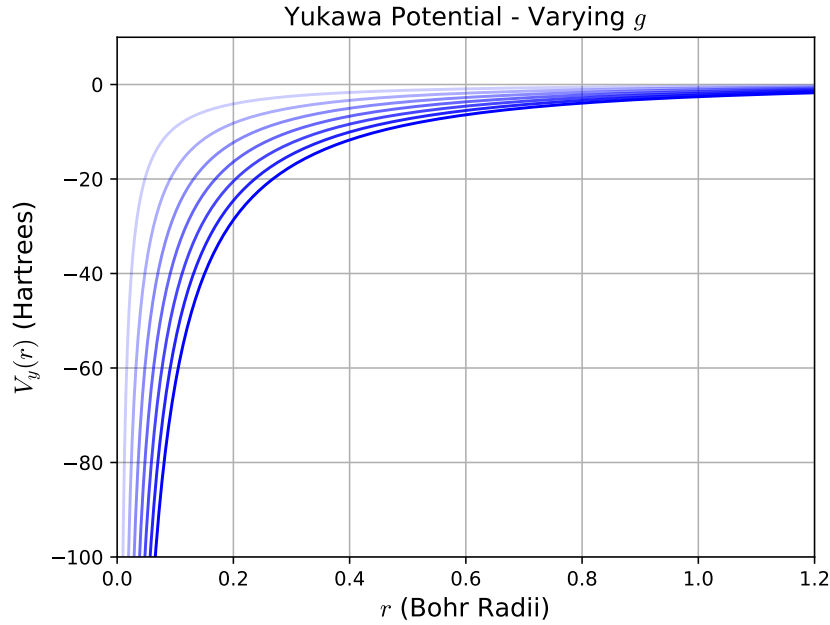


Figure 2.8: The Yukawa potential (Equation 2.37) for various values of g , the darkest blue curve corresponds to $g = -7$ and the lightest blue curve corresponds to $g = -1$, where g has been incremented by -1 for each darker shade starting from the lightest.

A disadvantage alluded to at the end of Section 2.4 is that, for the Yukawa potential, the phase shifts and scattering length do not have exact analytical solutions. Although the scattering length can be calculated using the first Born approximation [6], it is only reliable for small values of $|V_0|$, and becomes inaccurate for large values of $|V_0|$. As a consequence, it is not suitable for comparison with Numerov's method for large values of $|V_0|$.

The scattering length calculated using Numerov's method is showcased in Figure 2.9. For this calculation, points r_p and r_q had to be taken from the wave solution in the asymptotic region. Due to the Yukawa potential (2.37) asymptotically approaching zero, the asymptotic region was implemented with the cutoff: $V_y(r) < 1 \times 10^{-8}$. When the potential was in this region r_p and r_q values were selected. Additionally, to adjust the Yukawa potential the parameter g was varied, to make the potential attractive similar to what is shown in Figure 2.9, g values were made negative.

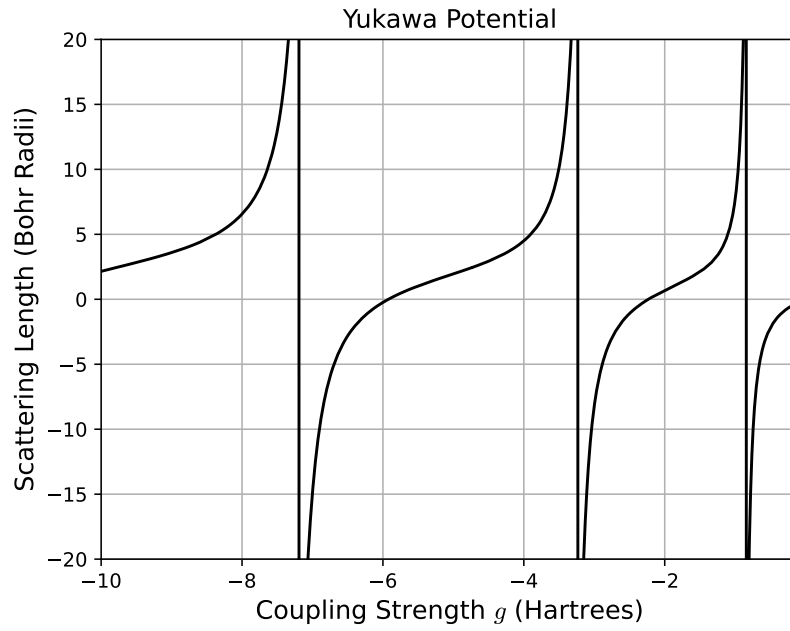


Figure 2.9: This figure shows the scattering length for the Yukawa potential where it was calculated using Numerov's method. The parameters used in the system were as follows: $E = 1 \times 10^{-3}$, $l = 0$, $h = 0.00001$, $\mu = 1$

Chapter 3

Metropolis-Hastings

In this chapter, we discuss the “backbone” of the diagrammatic Monte Carlo (DiagMC) method: the Metropolis-Hastings algorithm (MHA). The MHA is a Markov chain Monte Carlo (MCMC) method that generates samples from a target probability distribution. Samples from the probability distribution are proposed and accepted based on detailed balance equations that depend on the probability distribution and on the current sample, rather than the history of previous states (Markov chain). This chapter introduces Markov chains and the Metropolis-Hastings algorithm, applies them to the Maxwell-Boltzmann distribution, and highlights a diagrammatic Monte Carlo (DiagMC) approach to address issues that can arise. This will act as a soft introduction to DiagMC before it is introduced fully in Chapter 4.

3.1 Markov Chains

Suppose we have a system that is discretised in time: $t = 0, 1, 2, \dots$. At each time step a stochastic process is performed where the outcome determines the state of the system at that specific time, where we denote the state space as $S = \{x_0, x_1, x_2, \dots\}$. We also denote the system existing in state x at time t as $X_t = x$, where X_t is a random variable whose value gives the state of the system at time t , and has a probability distribution of

$$\{P(X_t = x) \mid x \in S\}. \quad (3.1)$$

In general, the next state, produced after a stochastic process may depend on the entire history of states i.e.

$$P(X_t = x \mid X_0 = x_0, X_1 = x_1, \dots, X_{t-1} = x_{t-1}). \quad (3.2)$$

This can be thought of as the system maintaining memory of the previous states. However, for a Markov chain, the next state produced is dependent solely on the current state

$$P(X_t = x \mid X_0 = x_0, X_1 = x_1, \dots, X_{t-1} = x_{t-1}) = P(X_t = x \mid X_{t-1} = x_{t-1}). \quad (3.3)$$

A system that employs a Markov process, can be considered memoryless, as it does not require information about any of the previous states except the current one, to generate the next state.

Consider the probability of transitioning to a given state. For a system that is transitioning from state X_{t-1} to X_t , the variable X_t can take any value in the state space: $X_t \in S$. Suppose we want to transition from state x_i to x_j for $x_i, x_j \in S$ (going forward we will refer to state $x_i \equiv i$ and $x_j \equiv j$), it is assumed that each transition from state i to j has an associated probability

$$P(X_t = j \mid X_{t-1} = i) = p_{ij} \quad \forall i, j \in S, \quad (3.4)$$

with the condition that $\forall i, j$

$$p_{ij} \geq 0 \quad \text{and} \quad \sum_j p_{ij} = 1. \quad (3.5)$$

This allows us to represent transition probabilities as a matrix $\mathbf{P}$

$$\mathbf{P} = \begin{pmatrix} p_{00} & p_{01} & p_{02} & \cdots \\ p_{10} & p_{11} & p_{12} & \cdots \\ p_{20} & p_{21} & p_{22} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \quad (3.6)$$

Given a transition probability matrix, a transition diagram can be drawn. Suppose we have a transition probability matrix

$$\mathbf{P}_3 = \begin{pmatrix} 0.4 & 0.4 & 0.2 \\ 0.1 & 0.9 & 0 \\ 0.2 & 0.5 & 0.3 \end{pmatrix}, \quad (3.7)$$

then we can build a transition diagram using this matrix ($\mathbf{P}_3$), illustrated in Figure 3.1.

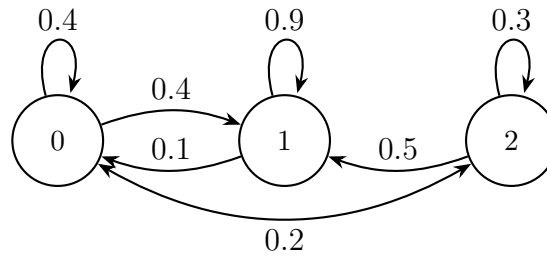


Figure 3.1: Transition diagram corresponding to the transition matrix $\mathbf{P}_3$ (Equation 3.7).

Now moving the analysis to the idea of many Markov chains. First, a finite, irreducible¹, aperiodic² Markov chain is examined, with states $0, 1, \dots, M$ such that $i, j \in 0, 1, \dots, M$. If we

¹A Markov chain that is fully connected. Within a finite number of steps you can visit any state, starting from any state.

²A Markov chain where a state can be revisited at irregular times. Furthermore, there is no periodic time where a state must be revisited.

start at state i , and want to examine what happens to the probability after transitioning n times, this is

$$p_{ij}^{(n)} \xrightarrow{n \rightarrow \infty} \pi_j \quad \forall i, j \quad \text{or} \quad \lim_{n \rightarrow \infty} \left[P(X_n = j \mid X_0 = i) \right] = \pi_j. \quad (3.8)$$

The probability after n transitions converges to a limiting value π_j as $n \rightarrow \infty$, where the initial state was i . This is equivalent to

$$\mathbf{P}^n \xrightarrow{n \rightarrow \infty} \begin{pmatrix} \pi_0 & \pi_1 & \dots & \pi_M \\ \pi_0 & \pi_1 & \dots & \pi_M \\ \vdots & \vdots & \ddots & \vdots \\ \pi_0 & \pi_1 & \dots & \pi_M \end{pmatrix}. \quad (3.9)$$

These probabilities $\{\pi_j \mid j = 0, 1, \dots, M\}$ are known as the stationary or limiting probabilities. Each row of the transition probability matrix in Equation 3.9 converges to the same row vector, which we define as $\boldsymbol{\pi}$, explicitly stated as: $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots, \pi_M)$, from this we obtain the following important relationship

$$\boldsymbol{\pi} = \boldsymbol{\pi} \mathbf{P} \quad (3.10)$$

where $\boldsymbol{\pi}$ is known as the stationary distribution. This is a distribution that, in the limit as $n \rightarrow \infty$, the system converges to. In the same limit, applying the transition probability matrix to the stationary distribution results in the same distribution. Therefore, for a finite, irreducible, and aperiodic Markov chain there exists a unique stationary distribution that is independent of the initial state ($\exists! \boldsymbol{\pi}$ such that $\boldsymbol{\pi} = \boldsymbol{\pi} \mathbf{P}$).

This idea of a stationary distribution is crucial in the context of the Metropolis-Hastings algorithm (MHA), as it constructs Markov chains whose stationary distribution is a specified target distribution. Allowing us to sample from distributions that may otherwise be difficult to sample.

3.2 Metropolis-Hastings Algorithm

This section covers a detailed description of the method behind the Metropolis-Hastings algorithm (MHA). Followed by an example of how the MHA is used in practice.

Consider a non-negative function $f(x)$ defined on the domain $x \in (0, \infty)$, that has the well defined integral

$$C = \int_0^\infty f(x) dx < \infty. \quad (3.11)$$

Suppose we wish to construct a Markov chain $\{x_n \mid x_n \in (0, \infty)\}$ where $n = 1, 2, 3, \dots$ whose stationary distribution $p(x)$ is given by

$$p(x) = \frac{f(x)}{C}. \quad (3.12)$$

The goal is to generate samples from $p(x)^3$ to be able to approximate the value of the following integral for a given function $A(x)$ using $p(x)$

$$I = \int_0^\infty A(x)f(x)dx. \quad (3.13)$$

We will see the significance of this integral when diagrammatic Monte Carlo in Chapter 4.2 is discussed. From this we will be able to calculate the expectation value of $A(x)$

$$\langle A \rangle = \frac{1}{C} \int_0^\infty A(x)f(x)dx \quad (3.14)$$

without requiring explicit knowledge of C . We achieve this by constructing a Markov chain, where the next state is accepted or rejected based on an associative probability, constructed to ensure that given enough steps, states will be sampled from the target distribution.

3.2.1 Method

We begin by motivating the method with the idea of a balance equation that governs the Markov process, and the justification for it. Consider N independent Markov chains, and define the i th ensemble as the collection of states across all N Markov chains at step i (shown in Figure 3.2). Let $N_a^{(i)}$ denote the number of times state a appears in ensemble i , as $N \rightarrow \infty$, we can define the probability of state a as [8]

$$\lim_{N \rightarrow \infty} \left(\frac{N_a^{(i)}}{N} \right) = W_a. \quad (3.15)$$

In the limit as $N \rightarrow \infty$, this defines the distribution for a given step, while the limit $i \rightarrow \infty$ ensures convergence to the stationary distribution. For a particular chain, we denote the probability of transitioning from state a at step i to state b at step $i + 1$ as $P_{a \rightarrow b}$. The goal is to define this probability ($P_{a \rightarrow b}$) such that in the statistical limit, the desired distribution is equal to W_a for all states. In the limit $i \rightarrow \infty$, the Markov chains reach the stationary distribution. In this regime, taking $N \rightarrow \infty$, the number $N_a^{(i)}$ becomes independent of i . This is conveyed in Figure 3.2, where advancing the stationary distribution by one step produces an ensemble that is distributed according to the stationary distribution. This allows us to relate the stochastic probability $P_{a \rightarrow b}$ and W_a with the condition

$$\langle N_a^{(i+1)} - N_a^{(i)} \rangle = 0. \quad (3.16)$$

If the i th ensemble is in the statistical limit, then the number of states that are a in ensemble i that change to state b due to the stochastic process to get ensemble $i + 1$ (departures) is equal to the number of states that transitioned from state b in i to state a at $i + 1$ (arrivals)

$$N_a^{(i+1)} - N_a^{(i)} = \sum_b A_{b \rightarrow a} - \sum_b D_{a \rightarrow b}, \quad (3.17)$$

³The Markov chain has an exact stationary distribution. However, in practice the samples obtained are approximately distributed according to $p(x)$.

where $A_{b \rightarrow a}$ is the number of transitions from state b to state a , and $D_{a \rightarrow b}$ denotes the number of transitions from state a to state b in one step. The expectation value of Equation 3.17 is found by representing $\langle A \rangle$ and $\langle D \rangle$ in the following ways

$$\langle A \rangle = N_b^{(i)} P_{b \rightarrow a} = N W_b P_{b \rightarrow a} \quad (3.18)$$

$$\langle D \rangle = N_a^{(i)} P_{a \rightarrow b} = N W_a P_{a \rightarrow b}. \quad (3.19)$$

The subtraction of Equation 3.19 from Equation 3.18 leads to the balance equation

$$\sum_b N W_b P_{b \rightarrow a} - \sum_b N W_a P_{a \rightarrow b} = 0 \quad \implies \quad \sum_b W_a P_{a \rightarrow b} = \sum_b W_b P_{b \rightarrow a}. \quad (3.20)$$

A sufficient condition for Equation 3.20 is the detailed balance equation, where we are now in the position to begin our discussion of the MHA

$$W_a P_{a \rightarrow b} = W_b P_{b \rightarrow a}. \quad (3.21)$$

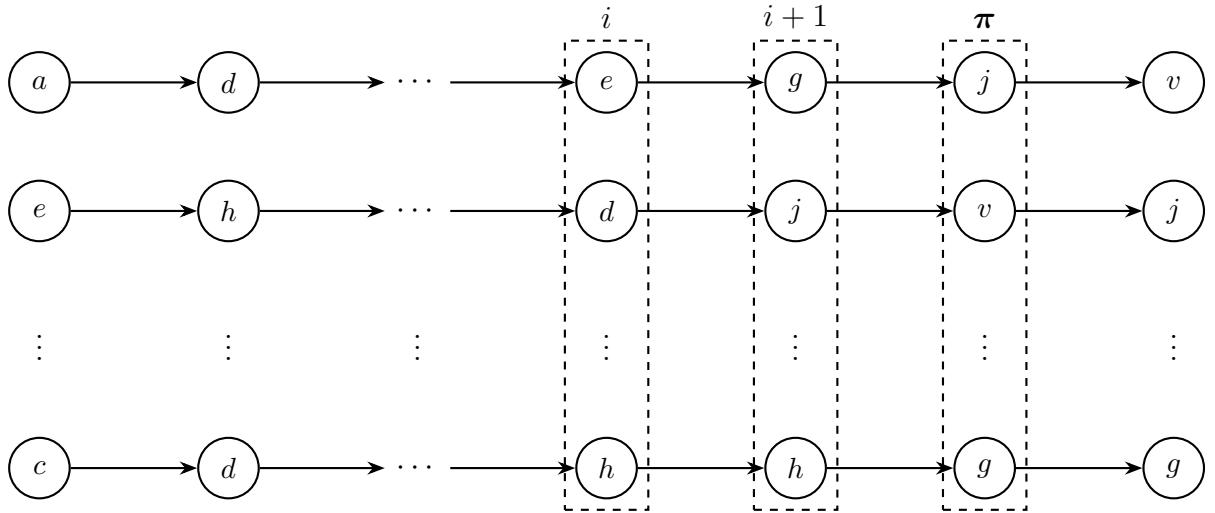


Figure 3.2: This illustration shows N Markov chains, where the possible allowed states are $a, b, c, d, e, f, \dots$. The dashed boxes around the i th elements are called the i th ensemble, similar for $i + 1$. The other dashed box labeled π is the stationary distribution for this theoretical system. Notice how the next evolution or step in states results in the stationary distribution again, although the first two lines have changed from $j \rightarrow v$ and $v \rightarrow j$, the ensemble viewed holistically shows all the states are exactly same as the previous ensemble (the stationary distribution ensemble).

We begin by discussing how probabilities such as $P_{a \rightarrow b}$ are treated in MHA. These probabilities are split into two parts: a proposal probability $P_{a \rightarrow b}^{(\text{propose})}$ and an acceptance probability $P_{a \rightarrow b}^{(\text{accept})}$, written as

$$P_{a \rightarrow b} = P_{a \rightarrow b}^{(\text{propose})} \cdot P_{a \rightarrow b}^{(\text{accept})}. \quad (3.22)$$

Suppose we have a Markov chain in state a at step i , and we propose to transition to state b at step $i + 1$ with the probability $P_{a \rightarrow b}^{(\text{propose})}$. We either accept or reject this proposal with

probability $P_{a \rightarrow b}^{(\text{accept})}$, if rejected, the state stays in state a for step $i + 1$. If the proposal is accepted, the state transitions to state b at step $i + 1$. Relating this back to our original problem, we express the stationary distribution we wish to determine in terms of $f(x)$ as,

$$W_a = p(x_a) = \frac{f(x_a)}{C}. \quad (3.23)$$

Substituting this equation into the detailed balance equation (3.21), the normalisation constant C cancels, allowing us to write the detailed balance equation as

$$f(x_a)P_{a \rightarrow b} = f(x_b)P_{b \rightarrow a}. \quad (3.24)$$

The probability $P_{a \rightarrow b}^{(\text{propose})}$ is written in terms of a probability density $\Omega(x, y)$, where y is a proposal state that is generated, and x is the current state. Moreover, this probability density has the condition

$$\int_0^\infty \Omega(x, y) dy = 1. \quad (3.25)$$

The probability $P_{a \rightarrow b}^{(\text{propose})}$ is explicitly written in terms of the probability density $\Omega(x, y)$ as

$$P_{a \rightarrow b}^{(\text{propose})} = \Omega(x_a, x_b) \Delta x \quad (3.26)$$

where Δx appears from discretising the interval $(0, \infty)$ into infinitesimally small lengths. Using the definition in Equation 3.26, we then rewrite the detailed balance equation as

$$f(x_a)\Omega(x_a, x_b)P_{a \rightarrow b}^{(\text{accept})} = f(x_b)\Omega(x_b, x_a)P_{b \rightarrow a}^{(\text{accept})} \quad (3.27)$$

where the Δx term has cancelled on either side, and we have defined $P_{a \rightarrow b}^{(\text{accept})}$ as [3]

$$P_{a \rightarrow b}^{(\text{accept})} = \begin{cases} 1 & \text{if } R(x_a, x_b) \geq 1 \\ R(x_a, x_b) & \text{if } R(x_a, x_b) < 1 \end{cases} \quad (3.28)$$

where $R(x_a, x_b)$ is the acceptance ratio [3], expressed by rearranging Equation 3.27 for $P_{a \rightarrow b}^{(\text{accept})}/P_{b \rightarrow a}^{(\text{accept})}$

$$R(x_a, x_b) = \frac{f(x_b)\Omega(x_b, x_a)}{f(x_a)\Omega(x_a, x_b)}. \quad (3.29)$$

In practice we have the ability to choose how the proposed states are sampled i.e. we have freedom of choice for $\Omega(x_a, x_b)$, as long as the condition from Equation 3.25 is satisfied. We discuss the choice of $\Omega(x_a, x_b)$ in more detail in Section 3.2.2.

The following is a structured recursive procedure that follows the Metropolis-Hastings algorithm to generate a Markov chain that, in the statistical limit, will converge to the target distribution.

1. Choose an initial state $x_0 \in (0, \infty)$
2. For $n = 0, 1, 2, \dots$
 - Given the current state x_n generate a proposal value x_{prop} using $\Omega(x_n, x_{prop})$
 - Compute the acceptance ratio
$$R = \frac{f(x_{prop})\Omega(x_{prop}, x_n)}{f(x_n)\Omega(x_n, x_{prop})}$$
 - Draw a random number r uniformly on $[0, 1]$
 - If $r \leq \min(1, R)$ accept the proposal:
$$x_{n+1} = x_{prop},$$

otherwise reject:

$$x_{n+1} = x_n.$$
3. Return the generated Markov chain:
$$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots$$

We now proceed by calculating the expectation value of a given function $A(x)$ using $p(x)$. The expectation value for $A(x)$ can be written for a Markov chain that is classed as ergodic (being aperiodic and irreducible) as [9]

$$\langle A \rangle = \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{n=1}^M A(x_n), \quad (3.30)$$

therefore, we can find the solution to I by multiplying Equation 3.30 by C

$$I = \int_0^\infty A(x)f(x)dx = C \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{n=1}^M A(x_n). \quad (3.31)$$

We can then calculate C from a “special case” of 3.31 [3], when $A(x) = g_0(x)/f(x)$, where the integral $\int_0^\infty g_0(x)dx$ can be performed analytically. Substituting in this special case to Equation 3.31, and rearranging for C we get

$$C = \int_0^\infty g_0(x)dx \left[\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{n=1}^M \frac{g_0(x_n)}{f(x_n)} \right]^{-1}, \quad (3.32)$$

combining Equation 3.32 and Equation 3.31 yields

$$\int_0^\infty A(x)f(x)dx = \int_0^\infty g_0(x)dx \lim_{M \rightarrow \infty} \left(\frac{\sum_{n=1}^M A(x_n)}{\sum_{n=1}^M g_0(x_n)/f(x_n)} \right). \quad (3.33)$$

Equation 3.33 allows us to evaluate the integral I , purely by building a Markov chain that samples from a distribution based on $f(x)$.

3.2.2 Metropolis-Hastings Results

We now look at an example where we build a Markov chain towards the target distribution based on the Maxwell-Boltzmann Distribution $f_{MB}(v)$ [10],

$$f_{MB}(v) = \left[\frac{m}{2\pi k_B T} \right]^{3/2} 4\pi v^2 \exp \left(-\frac{mv^2}{2k_B T} \right), \quad (3.34)$$

where k_B is the Boltzmann constant, T is temperature (in Kelvin), and m is mass (in kg). The goal is to build a Markov chain towards the target distribution based on $f_{MB}(v)/C$. We start by constructing the Markov chain described in Section 3.2.1, in practice we create one long chain rather than multiple independent chains. A typical chain calculated is seen in Figure 3.3

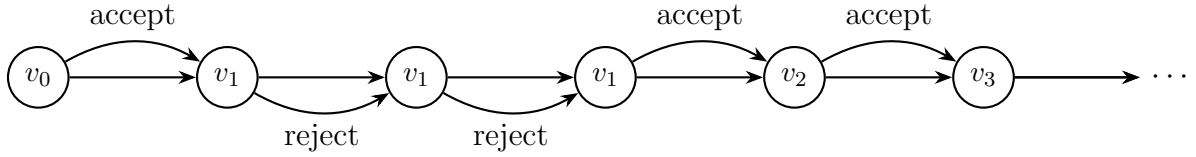


Figure 3.3: The Markov chain is initialised with velocity v_0 . The velocity v_1 is then proposed and accepted, this is followed by two proposal rejections, resulting in the same velocity v_1 being recorded in the Markov chain.

The velocity axis was partitioned into bins and a Markov chain was generated where each sample or velocity from the Markov chain was distributed into their respective bin. This was then used to construct a histogram seen in Figure 3.4. The choice of $\Omega(v_a, v_b)$ was chosen such that the proposal values were uniformly selected within the interval $(\eta^{-1}v_a, \eta v_a)$, where η was a constant chosen to either expand or diminish the “window” size for $\eta > 1$

$$\Omega(v_a, v_b) = \begin{cases} \frac{\eta}{(\eta^2-1)v_a} & \text{if } v_b \in (\eta^{-1}v_a, \eta v_a), \\ 0 & \text{otherwise.} \end{cases} \quad (3.35)$$

The choice of this particular $\Omega(v_a, v_b)$ was made to simplify the acceptance ratio to

$$R(v_a, v_b) = \frac{v_a f_{MB}(v_b)}{v_b f_{MB}(v_a)} \quad (3.36)$$

where the constant η cancels.

Results for the calculation of I (Equation 3.13), using the same $\Omega(v_a, v_b)$ as above (Equation 3.35), are shown by calculating $\langle v \rangle$ and C separately, this was done using Equations 3.30 and 3.32 respectively. By multiplying the two calculations together one can show the calculation of I (Equation 3.31). Figure 3.5 shows both $\langle v \rangle$ and C calculations, that were computed using $g_0(x)$, which is expressed as

$$g_0(x) = \begin{cases} \frac{1}{A_r} & \text{if } 0 \leq x \leq A_r \\ 0 & \text{otherwise,} \end{cases} \quad (3.37)$$

where A_r was a constant chosen as 2×10^5 . From 3.37 it is shown that the integral is evaluated as

$$\int_0^\infty g_0(x) dx = \int_0^{A_r} \frac{1}{A_r} dx = 1. \quad (3.38)$$

The results seen in Figure 3.5 show that as the number of steps increases, we converge to our intended values.

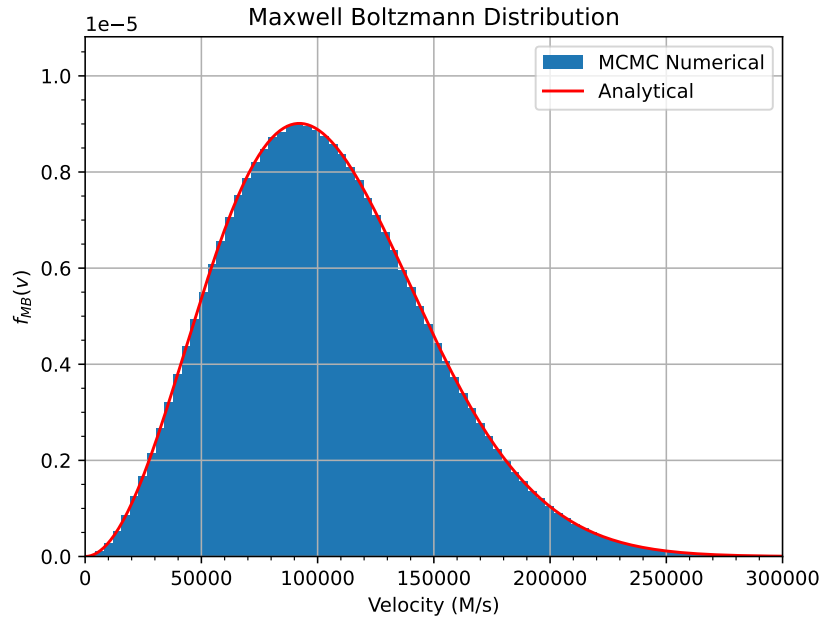


Figure 3.4: This is a normalised histogram approximation to the Maxwell-Boltzmann distribution (analytical in red) using Markov chain Monte Carlo sampling, that includes 100 bins, a temperature of $T = 280\text{K}$, an initialised value $v_0 = \frac{2}{\sqrt{\pi}} \sqrt{\frac{2k_B T}{m}}$, where m was the mass of an electron and k_B was the Boltzmann constant both implemented using the SciPy Python package [11] using their normal SI units. The amount of steps in the Markov chain was $N = 1 \times 10^7$ and η from Equation 3.35 was set equal to 1.5

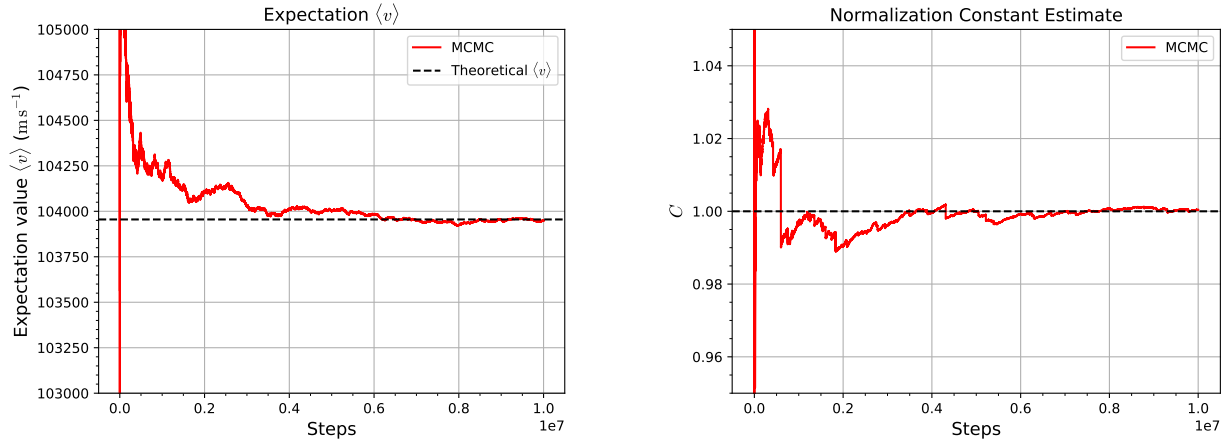


Figure 3.5: Metropolis-Hastings algorithm results for the calculation of the expectation value of the velocity for the Maxwell-Boltzmann distribution and the normalisation constant, these were calculated using Equations 3.30 and 3.32 respectively. The parameters of both systems were as follows: $T = 280\text{K}$, $\eta = 2$ (from Equation 3.35), the number of Monte Carlo steps was $N = 1 \times 10^7$, the initial v_0 in the chain was initialised to $v_0 = \frac{2}{\sqrt{\pi}} \sqrt{\frac{2k_B T}{m}}$, where m and k_B are the mass of an electron and Boltzmann constant respectively [11]. Finally, both graphs used $g_0(x)$ (from Equation 3.37) in their calculations, where A_r was set equal to 2×10^5 .

3.3 Diagrammatic Monte Carlo Trick

A problem can arise in Equation 3.33 (Section 3.2.1), in cases where $f(x)$ becomes very small, the denominator: $\sum_{n=1}^M g_0(x_n)/f(x_n)$ can become problematic, leading to noisy approximations of the integral I in Equation 3.33. We can remedy this by approaching the problem using diagrammatic Monte Carlo (DiagMC). We first enlarge the configuration space or state space⁴ by introducing two “Types”. Type-1 corresponds to updating the Markov chain with proposal states and accepting or rejecting them like normal. Whereas Type-0 is an auxiliary configuration that contributes nothing **directly** to the quantity being estimated, however its purpose is for normalisation, as we will see in the following analysis. This extended configuration space attempts to construct a new Markov chain based on weights associated with both types rather than $f(x)/C$. The total statistical weight of the system is now

$$D_0 + \int_0^\infty f(x)dx = D_0 + C \quad (3.39)$$

where D_0 is the weight given to Type-0. The probability of the system occupying Type-0 is then

$$p_0 = \frac{D_0}{D_0 + C}, \quad (3.40)$$

and the probability of finding the system in Type-1 in the interval $[x, x + dx]$ is

$$dp(x) = \frac{f(x)dx}{D_0 + C}. \quad (3.41)$$

⁴The configuration space, up until this point has been the allowed states that x_n can take: $x_n \in (0, \infty)$ for $n = 1, 2, 3, \dots$

We then define $\sigma_n^{(\nu)}$, where ν corresponds to the type and can equal either 0 (Type-0) or 1 (Type-1). We then assign the value to $\sigma_n^{(\nu)} = 1$ if the n th state in the Markov chain is in Type- ν and zero otherwise. These act as the number of steps that the Markov chain was in a specific Type for. Since Type-1 only contributes to the estimator, we introduce A_n such that

$$A_n = \begin{cases} A(x_n) & \text{if } \nu = 1 \text{ System is in Type-1} \\ 0 & \text{if } \nu = 0 \text{ System is in Type-0} \end{cases} \quad (3.42)$$

Allowing us to define the following sums

$$S_A(M) = \sum_{n=1}^M A_n \quad (3.43)$$

$$S_\nu(M) = \sum_{n=1}^M \sigma_n^{(\nu)}. \quad (3.44)$$

Using Equations 3.43 and 3.44 provide us an alternative way of calculating the expectation value in the statistical limit

$$\langle A \rangle = \lim_{M \rightarrow \infty} \frac{S_A(M)}{S_1(M)} = \frac{1}{C} \int_0^\infty A(x) f(x) dx \quad (3.45)$$

The probability of the system occupying Type-0 and Type-1 can be related to find expressions for C and D_0 . The probability of the system being in Type-1, using Equation 3.39 and 3.44, is

$$\frac{S_1(M)}{M} \xrightarrow{M \rightarrow \infty} p_1 = \frac{C}{D_0 + C}. \quad (3.46)$$

The probability for Type-0, using Equation 3.40 and 3.44, is then

$$\frac{S_0(M)}{M} \xrightarrow{M \rightarrow \infty} p_0 = \frac{D_0}{D_0 + C}. \quad (3.47)$$

Taking the ratio of Equation 3.46 and 3.47 in the limit $M \rightarrow \infty$, gives us

$$\lim_{M \rightarrow \infty} \frac{S_1(M)}{S_0(M)} = \frac{C}{D_0}. \quad (3.48)$$

This gives us a way to calculate C and I using only the number of steps in each Type and Equation 3.43

$$C = \int_0^\infty f(x) dx = D_0 \lim_{M \rightarrow \infty} \frac{S_1(M)}{S_0(M)} \quad (3.49)$$

substituting C into Equation 3.45 yields

$$I = \int_0^\infty A(x) f(x) dx = D_0 \lim_{M \rightarrow \infty} \frac{S_A(M)}{S_0(M)}. \quad (3.50)$$

As discussed to produce a Markov chain that supports this expanded configuration space, we pair our original configuration of proposing states and either accepting or rejecting them, with

the proposal of switching between Types. For example, if you are in Type-1, the system would propose to swap to Type-0, just like it would propose a state. This Type transfer also has an acceptance ratio associated with it, that we will discuss how to find now.

We seek to construct a detailed balance equation for the transition between Types, to do this we first describe how transitions are constructed.

1. **Type 0 $\rightarrow$ 1:** We propose this transition with probability $p_{0 \rightarrow 1}^{(\text{propose})}$. If the system is already in Type-1, this transition is rejected immediately, otherwise we propose a Type-1 configuration by generating a sample x_a from distribution function $\Omega(x)$. The proposal is accepted with probability $P_{0 \rightarrow 1}^{(\text{accept})}(x_a)$.
2. **Type 1 $\rightarrow$ 0:** If the system is already in Type-0 reject the transfer. Otherwise, propose this transition with probability $p_{1 \rightarrow 0}^{(\text{propose})}$. Accept this proposal with probability $P_{1 \rightarrow 0}^{(\text{accept})}(x_a)$.

Therefore, the DBE (derived in Appendix F) for this system is

$$f(x_a) p_{1 \rightarrow 0}^{(\text{propose})} P_{1 \rightarrow 0}^{(\text{accept})}(x_a) = D_0 p_{0 \rightarrow 1}^{(\text{propose})} \Omega(x_a) P_{0 \rightarrow 1}^{(\text{accept})}(x_a) \quad (3.51)$$

where $p_{1 \rightarrow 0}^{(\text{propose})}$ is the proposal probability of transferring from Type-1 to Type-0, $P_{1 \rightarrow 0}^{(\text{accept})}$ is the acceptance probability of transferring from Type-1 to Type-0. We define the acceptance probabilities as

$$P_{0 \rightarrow 1}^{(\text{accept})}(x) = \begin{cases} 1 & \text{if } R_{0 \rightarrow 1}(x) \geq 1 \\ R_{0 \rightarrow 1}(x) & \text{if } R_{0 \rightarrow 1}(x) < 1 \end{cases} \quad (3.52)$$

$$P_{1 \rightarrow 0}^{(\text{accept})}(x) = \begin{cases} 1 & \text{if } R_{1 \rightarrow 0}(x) \geq 1 \\ R_{1 \rightarrow 0}(x) & \text{if } R_{1 \rightarrow 0}(x) < 1 \end{cases} \quad (3.53)$$

$$(3.54)$$

where $R_{0 \rightarrow 1}$ is the acceptance ratio to transfer from Type-0 to Type-1. This is found by rearranging Equation 3.51 to solve for $P_{0 \rightarrow 1}^{(\text{accept})}(x_a)/P_{1 \rightarrow 0}^{(\text{accept})}(x_a)$

$$R_{0 \rightarrow 1}(x) = \frac{p_{1 \rightarrow 0}^{(\text{propose})}}{p_{0 \rightarrow 1}^{(\text{propose})}} \frac{f(x)}{D_0 \Omega(x)}, \quad (3.55)$$

from this we find

$$R_{1 \rightarrow 0}(x) = \frac{1}{R_{0 \rightarrow 1}(x)}. \quad (3.56)$$

From this we have all the ingredients to solve the problem stated at the beginning of Section 3.2.

By reformatting the configuration space into two Types we have shown that we can evade the problem (discussed in the beginning of this section) by splitting the contributions of the original configuration (described in Section 3.2.1) into a single type (Type-1), while Type-0 allows us to estimate the constant without having to sample from regions where $f(x)$ becomes small. The introduction of these types has effectively reformatted the problem in calculating the normalisation constant C into the ratio between the amount of steps in each type.

3.3.1 DiagMC Trick Results

We show calculations of $\langle v \rangle$ and C for the Maxwell-Boltzmann distribution similar to Chapter 3.2.2, where we can easily show the calculation of I (Equation 3.13) given these two ingredients. However, now we use the method discussed in Section 3.3. To produce the plots shown in Figure 3.6, the function $\Omega(v)$ (not dependent on proposed state) is used within the acceptance ratio in Equation 3.55, and was chosen to be proportional to the Heaviside function [3], defined as

$$\Theta(x_0 - x) = \begin{cases} 1 & \text{if } x < x_0 \\ 0 & \text{if } x > x_0. \end{cases} \quad (3.57)$$

We then define $\Omega(v)$ using the Heaviside function from Equation 3.57

$$\Omega(v) = \frac{\Theta(A_r - v)}{A_r}. \quad (3.58)$$

where A_r takes on the same value (2×10^5) as seen in Chapter 3.2.2. This leads to the following acceptance ratio

$$R_{0 \rightarrow 1}(v) = \frac{p_{1 \rightarrow 0}^{(\text{propose})}}{p_{0 \rightarrow 1}^{(\text{propose})}} \frac{A_r f(v)}{D_0} \quad (3.59)$$

$$R_{1 \rightarrow 0}(v) = \Theta(A_r - v) \frac{p_{0 \rightarrow 1}^{(\text{propose})}}{p_{1 \rightarrow 0}^{(\text{propose})}} \frac{D_0}{A_r f(v)} \quad (3.60)$$

where D_0 was set to unity, $p_{1 \rightarrow 0}^{(\text{propose})}$ and $p_{0 \rightarrow 1}^{(\text{propose})}$ were both constants set to 0.5 and 1.0 respectively. The Figure 3.6 shows correct convergence as the number of steps increases, where both figures used 1×10^8 steps. The expectation value $\langle v \rangle$ and C were calculated using Equations 3.45 and 3.49 respectively.

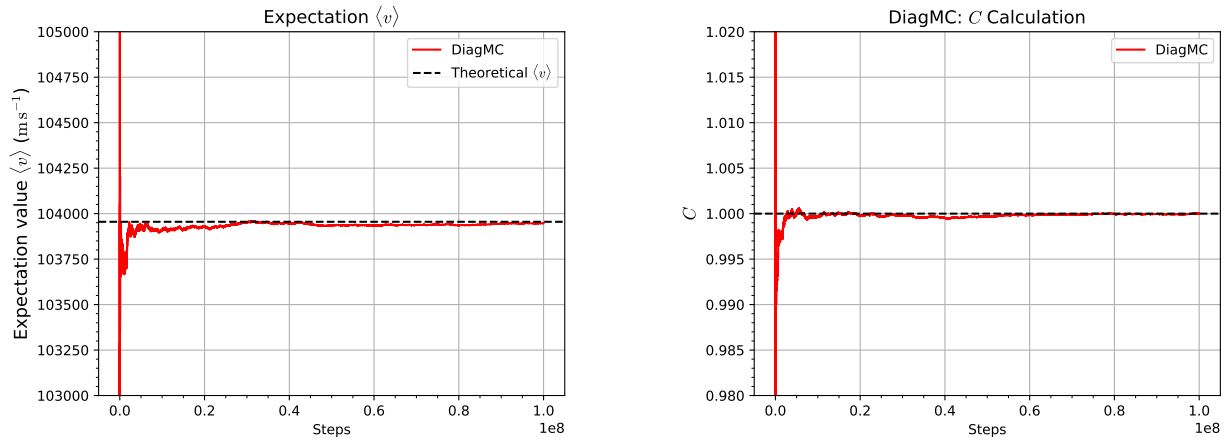


Figure 3.6: Calculated $\langle v \rangle$ and normalisation constant C from the Maxwell-Boltzmann distribution using the DiagMC method. These values were calculated using the following parameters: temperature as $T = 280\text{K}$, $\eta = 1.7$, the amount of Monte Carlo steps $N = 1 \times 10^8$, $A_r = 2 \times 10^5$ which was used within the Heaviside function (Equation 3.57). The initial v_0 in the chain was set equal to $v_0 = \frac{2}{\sqrt{\pi}} \sqrt{\frac{2k_B T}{m}}$, where k_B and m were the Boltzmann constant and mass of an electron, used in code from the SciPy Python package [11].

Chapter 4

Diagrammatic Monte Carlo

This chapter covers the motivation and method for the diagrammatic Monte Carlo (DiagMC) method. We then discuss a version of DiagMC called the bold line Diagrammatic Monte Carlo (BLDMC) [12] whereby limitations of DiagMC are alleviated. Afterwards we return to the scattering problem discussed in Chapter 2.1 and calculate the scattering length using BLDMC for two different systems, one governed by the square well potential (Equation 2.14) and another by the Yukawa potential (Equation 2.37). These are then compared with results obtained from Numerov's numerical method (Sections 2.4 and 2.5).

4.1 Motivation for Diagrammatic Monte Carlo

Diagrammatic Monte Carlo (DiagMC) is a stochastic method used for evaluating quantities expressed as diagrammatic expansions. Typically, DiagMC is performed with Feynman diagrams [13], however, they may also be used with propagators (Green's functions) and interaction potentials. For the latter two, we will see that by reformatting the scattering problem into its integral representation known as the Lippmann-Schwinger equation, we will be able to solve for the scattering length that uses both a propagator and interaction potential (Square well and Yukawa separately). A general diagrammatic expansion takes the form of the following equation [5]

$$Q(y) = \sum_{m=0}^{\infty} \sum_{\xi_m} \int \mathcal{D}(\xi_m, y, x_1, \dots, x_m) dx_1 \cdots dx_m \quad (4.1)$$

where $Q(y)$ is the quantity we wish to find, ξ_m is the index that corresponds to different terms with the same order m , for instance Feynman diagrams have first-order, second-order, etc. [1]. $x_1, x_2, \dots$ are the internal variables, and lastly, $\mathcal{D}$ denotes the contribution associated with a given diagram, including its statistical weight.

The general idea behind DiagMC is to approach a series of diagrams statistically rather than summing all terms explicitly. This is achieved by constructing a Markov chain that samples

diagrams with probabilities proportional to the diagram's weight, from this, $Q(y)$ is calculated through the accumulation of all sampled diagrams. As shown in our "soft" introduction to DiagMC in Chapter 3.3, the configuration space was expanded by introducing two "types". These types allowed the integral I , seen in Equation 3.13, to be estimated based on the frequency at which types were visited (Equation 3.50). In the general case, configuration spaces are expanded to be much larger, where each configuration would then correspond to a particular diagram specified by its topology order, weight, and internal variables for each diagram.

Up to this point, we have been discussing a general or standard version of DiagMC. However, this form has a limitation concerning the rapid growth with the amount of diagrams there are, as the order increases. Diagrams with opposite signs can cancel, making convergence difficult (illustrated in the top diagram on Figure 4.1). A variation on DiagMC: Bold Line Diagrammatic Monte Carlo (BLDMC) is introduced [5] and applied [12] in papers to overcome this problem. BLDMC reduces the amount of diagrams sampled from, by introducing self-referential or self-consistent expansions, seen in the bottom diagram in Figure 4.1 represented by the bold line. These bold lines are re-summations of diagrams, that effectively replace the expansion into a single diagrammatic object. In the problem presented in this paper, this bold line object appears through the usage of the Lippmann-Schwinger equation, where we will come to see the scattering amplitude is defined in terms of itself.

We use BLDMC rather than DiagMC to solve our particular problem as we re-frame the scattering problem as stated earlier to be represented by the Lippmann-Schwinger equation, which is a self-referential equation [6]

$$\psi(\mathbf{r}) = \phi(\mathbf{r}) + \int G_0(k, \mathbf{r}, \mathbf{r}') V(\mathbf{r}') \psi(\mathbf{r}') d\mathbf{r}', \quad (4.2)$$

where $\psi(\mathbf{r})$ is the wave function present on both sides of the equation, $\phi(\mathbf{r})$ is the incident particle, $G_0(k, \mathbf{r})$ is the Green's function that represents the propagation between scattering events (Figure 4.1), and finally $V(r)$ is the potential. From Equation 4.2 we find a solution to the scattering amplitude for the system, by determining the statistical weight that each configuration contributes through the Metropolis-Hastings process, where we reconstruct the scattering amplitude based on the accumulated measurements. Whereby this estimated distribution is used self-referentially to produce proposed values, in conjunction with acceptance ratios, these proposed values will be used to either accept or reject transfers to different diagrams.

By solving the scattering length in these scattering systems governed by the square well and Yukawa potential, we intend to show the usefulness and potential that the DiagMC or bold line Diagrammatic Monte Carlo method has in calculating diagrammatic expansions.

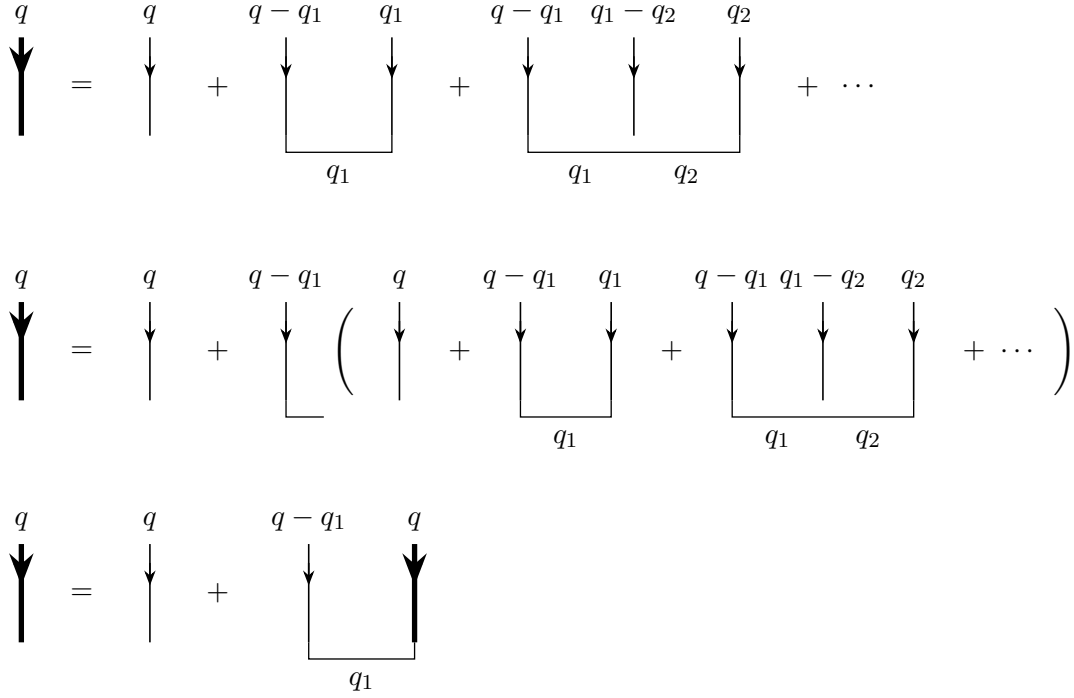


Figure 4.1: The top diagram shows a generic perturbative or diagrammatic expansion. Each term represents a diagram where each higher-order term introduces another variable (given as momentum $q, q_1, q_2, \dots$). The middle diagram shows the observation that by grouping terms in such a way, allows us to re-introduce the full expansion (top diagram) into the equation. This produces the bottom diagram which re-organises the sum into a single diagrammatic object that is used in a self-referential or Dyson-type expansion. Finally, q_1 and q_2 are the integration or internal variables and q is the external momentum.

4.2 Diagrammatic Monte Carlo Method

We begin this section from the integral representation of the Schrödinger equation (in a Lippmann-Schwinger type form), seen in Equation 4.2. We take it in the zero-energy case ($k \rightarrow 0$) and define the pseudo-potential $\tilde{U}$, an effective interaction that provides a simpler description of the scattering system, while maintaining the same underlying physics of the system

$$\psi(\mathbf{r}) = 1 + \int G_0(0, \mathbf{r}, \mathbf{r}') \tilde{U}(\mathbf{r}') d^3 \mathbf{r}' \quad (4.3)$$

$$\text{where} \quad \tilde{U}(\mathbf{r}) = V(\mathbf{r})\psi(\mathbf{r}). \quad (4.4)$$

Where ϕ represents the incident wave: e^{ikz} along the z axis and d^3 is introduced to remove ambiguity. In the limit $k \rightarrow 0$, the incident wave becomes one. We then multiply Equation 4.3 by $V(\mathbf{r})$ to find

$$\tilde{U}(\mathbf{r}) = V(\mathbf{r}) + V(\mathbf{r}) \int G_0(0, \mathbf{r}, \mathbf{r}') \tilde{U}(\mathbf{r}') d^3 \mathbf{r}', \quad (4.5)$$

and using the zero-energy free Green's function in real space [6] we get

$$\tilde{U}(\mathbf{r}) = V(\mathbf{r}) - V(\mathbf{r}) \int \frac{1}{2\pi|\mathbf{r} - \mathbf{r}'|} \tilde{U}(\mathbf{r}') d^3\mathbf{r}' \quad (4.6)$$

$$G_0(0, \mathbf{r}, \mathbf{r}') = -\frac{1}{2\pi|\mathbf{r} - \mathbf{r}'|} \quad \text{for } m = 1 \quad (4.7)$$

taking the Fourier transform into momentum space, we get

$$\tilde{U}(\mathbf{q}) = V(\mathbf{q}) - \int \frac{d^3q_1}{(2\pi)^3} V(\mathbf{q} - \mathbf{q}_1) \frac{2}{q_1^2} \tilde{U}(\mathbf{q}_1) \quad (4.8)$$

where we have also changed the integrating variable from $\mathbf{r}'$ to $\mathbf{q}_1$. Thus, we have derived the pseudo potential, where $G^{(0)}(q) = 2/q^2$ represents our propagator for a free particle. The scattering length is then found from the following equation [4]

$$\alpha = \frac{1}{2\pi} \tilde{U}(q = 0) \quad (4.9)$$

where $\hbar$ in the following analysis along with mass have both be set to unity. To be in a position to use BLDMC to find the scattering length, we first need to rescale the following variables

$$f(\mathbf{q}) = \frac{1}{2\pi} \tilde{U}(\mathbf{q}) \quad (4.10)$$

$$-u(\mathbf{q}) = \frac{1}{2\pi} V(\mathbf{q}) \quad (4.11)$$

substituting these into Equation 4.8 yields

$$f(\mathbf{q}) = -u(\mathbf{q}) - \int \frac{u(\mathbf{q} - \mathbf{q}_1) f(\mathbf{q}_1)}{2\pi^2 q_1^2} d^3q_1. \quad (4.12)$$

Since we will be looking at spherically symmetrical potentials only, we can then make the following simplification

$$f(q) = -u(q) - \frac{1}{\pi} \int_{-1}^1 d\chi \int_0^\infty u(|\mathbf{q} - \mathbf{q}_1|) f(q_1) dq_1 \quad (4.13)$$

where $\chi = \cos(\theta)$, and $|\mathbf{q} - \mathbf{q}_1| = \sqrt{q^2 + q_1^2 - 2qq_1\chi}$. Setting $q = 0$ in Equation 4.13 produces the following result

$$\alpha = u(0) - \frac{2}{\pi} \int_0^\infty u(q) f(q) dq. \quad (4.14)$$

Thus, we have set up the problem. However, we include one more addition, that being a parameter introduced by Prokof'ev et al [12] that will help the scattering system converge at points of resonance. We first express Equation 4.13 as

$$|f\rangle = |b\rangle + \mathbf{A}|f\rangle \quad (4.15)$$

$$\text{where } \langle q|b\rangle \equiv -u(q) \quad \langle q|f\rangle \equiv f(q) \quad (4.16)$$

$$\text{and } \langle q|\mathbf{A}|f\rangle = -\frac{1}{\pi} \int_{-1}^1 d\chi \int_0^\infty u(|\mathbf{q} - \mathbf{q}_1|) f(q_1) dq_1. \quad (4.17)$$

By subtracting the right-hand side of $|f\rangle = |b\rangle + \mathbf{A}|f\rangle$ from both sides we notice that

$$|f\rangle - |b\rangle - \mathbf{A}|f\rangle = 0 \quad (4.18)$$

and that we can multiply this by $\lambda\mathbf{A}$ and still get zero, where λ represents the tuning parameter

$$\lambda\mathbf{A}(|f\rangle - |b\rangle - \mathbf{A}|f\rangle) = 0, \quad (4.19)$$

we then add this to the right-hand side of Equation 4.15 to produce

$$|f\rangle = |b\rangle + (1 + \lambda)\mathbf{A}|f\rangle - \lambda\mathbf{A}|b\rangle - \lambda\mathbf{A}^2|f\rangle \quad (4.20)$$

now working in momentum representation we get

$$f = \underbrace{-u}_{\mathcal{A}} + \underbrace{(1 + \lambda)\mathbf{A}f}_{\mathcal{B}} + \underbrace{\lambda\mathbf{A}u}_{\mathcal{C}} + \underbrace{(-\lambda\mathbf{A}^2f)}_{\mathcal{D}}. \quad (4.21)$$

The parameter λ , as stated previously helps with convergence, this is proved and discussed more in Appendix G. The labels $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$ denote the different diagrams in the expanded configuration space associated with Equation 4.21. We perform the bold line diagrammatic Monte Carlo method on the configuration space, by performing stochastic summations over each diagram, as discussed in Chapters 3.3 and 4.1. The Monte Carlo updates are constructed such that configurations are sampled with probabilities proportional to the weight of each diagram. The weights of each diagram are found by taking the modulus of the specific diagram term in Equation 4.21. The updates are implemented with the Metropolis-Hastings regime, which generates a Markov chain, that satisfies the detailed balance equation constructed through the allowed transitions between diagrams. The crucial difference between the example of DiagMC seen in Chapter 3.3 and how we implement the bold line diagrammatic Monte Carlo method here, is the target distribution appears self-consistently to sample or propose momentum values. Rather than a fixed distribution, we have to build an approximation for what we are solving to sample from it. Another key difference is how the histogram is built. For each iteration in the algorithm, depending on the sign of the configuration that the system is left in, we either contribute $+1$ or -1 to the appropriate histogram bin (the current q at a given step). Similar to what was seen in Chapter 3.3, each diagram has the ability to transfer to another diagram with two associated probabilities, a proposal probability and acceptance probability, these probabilities along with which diagram transfers are possible allows us to define the acceptance ratios. In the next section we discuss the procedure and the allowed diagrammatic transitions. We then discuss how results are extracted in terms of approximated histograms, and finally how the scattering length is found.

4.3 Bold Line Diagrammatic Monte Carlo Procedure

This section has been split up into subsections containing diagrams and their allowed transfers. An illustration showing the possible diagram transfers can be seen in Figure 4.2, where the

probabilities between them are the addressing probabilities that were used when taking results (Section 4.5). The acceptance ratios (R) were constructed (Derived in Appendix H) by enlarging the configuration space, defining the transitions between each diagram and deriving the corresponding Metropolis-Hastings acceptance ratios by building the detailed balance equation based on these transfer updates.

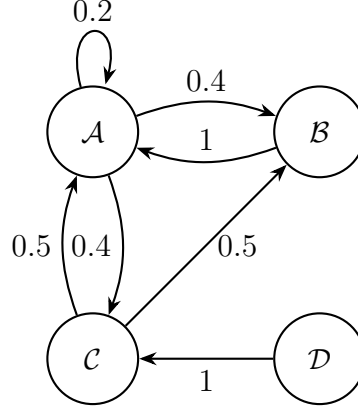


Figure 4.2: Transition schematic featuring all possible diagram transfers and the **proposal** probability associated with them.

4.3.1 Diagram $\mathcal{A}$

$\mathcal{A} \rightarrow \mathcal{A}$

The acceptance ratio for this transfer is given as unity: $R_{\mathcal{A} \rightarrow \mathcal{A}} = 1$, along with a new variable q in diagram $\mathcal{A}$ being generated from the normalised probability density

$$p(q) = \frac{|u(q)|}{I_u} \quad \text{where} \quad I_u = \int_0^\infty |u(q)| dq \quad (4.22)$$

how this value (q) is sampled from this distribution is discussed in Appendix I.

$\mathcal{A} \rightarrow \mathcal{B}$

Diagram $\mathcal{B}$ contains the integration variables q, q', χ , where q' is the proposed momentum value. These values are proposed by uniformly selecting from the interval $[-1, 1]$ for χ , and selecting q' from the probability distribution that we are generating $|f(q')|$. The acceptance ratio is

$$R_{\mathcal{A} \rightarrow \mathcal{B}} = \frac{2|1 + \lambda|I_f}{\pi p_{\mathcal{AB}}} \left| \frac{u(\mathbf{q} - \mathbf{q}')}{u(q)} \right| \quad (4.23)$$

where $p_{\mathcal{AB}}$ is the proposal probability of transferring from diagram $\mathcal{A} \rightarrow \mathcal{B}$, and I_f is

$$I_f = \int_0^\infty |f(q)| dq \quad (4.24)$$

this is proportional to the summation of all contributions to the histogram.

$\mathcal{A} \longrightarrow \mathcal{C}$

We propose values χ uniformly on the interval $[-1, 1]$, and select a proposal value for q' from the probability density $|u(q')|$, giving the acceptance ratio of

$$R_{\mathcal{A} \rightarrow \mathcal{C}} = \frac{2|\lambda|I_{upc\mathcal{A}}}{\pi p_{\mathcal{AC}}} \left| \frac{u(|\mathbf{q} - \mathbf{q}'|)}{u(q)} \right| \quad (4.25)$$

4.3.2 Diagram $\mathcal{B}$

$\mathcal{B} \longrightarrow \mathcal{A}$

For this update we only attempt to move back. Therefore, the acceptance ratio is

$$R_{\mathcal{B} \rightarrow \mathcal{A}} = \frac{1}{R_{\mathcal{A} \rightarrow \mathcal{B}}}, \quad (4.26)$$

where $R_{\mathcal{A} \rightarrow \mathcal{B}}$ is given in Equation 4.23

4.3.3 Diagram $\mathcal{C}$

$\mathcal{C} \longrightarrow \mathcal{A}$

For this transfer we only attempt to move back resulting in the acceptance ratio being expressed as

$$R_{\mathcal{C} \rightarrow \mathcal{A}} = \frac{1}{R_{\mathcal{A} \rightarrow \mathcal{C}}}, \quad (4.27)$$

where $R_{\mathcal{A} \rightarrow \mathcal{C}}$ is given by Equation H.6

$\mathcal{C} \longrightarrow \mathcal{D}$

Here, we propose χ' uniformly in the interval $[-1, 1]$ and q'' from the probability distribution $|f(q'')|$. An important note is that, we do not resample q' or χ . Thus, the acceptance ratio is given as

$$R_{\mathcal{C} \rightarrow \mathcal{D}} = \frac{2I_f}{\pi p_{\mathcal{CD}}} \left| \frac{u(|\mathbf{q}' - \mathbf{q}''|)}{u(q')} \right| \quad (4.28)$$

4.3.4 Diagram $\mathcal{D}$

$\mathcal{D} \longrightarrow \mathcal{C}$

Lastly, we do not sample any values and only try to transfer back. As a consequence the acceptance ratio is

$$R_{\mathcal{D} \rightarrow \mathcal{C}} = \frac{1}{R_{\mathcal{C} \rightarrow \mathcal{D}}}. \quad (4.29)$$

4.4 Calculating Scattering Length with BLDMC

As previously mentioned histograms generated would be produced from $+1$ or -1 contributions depending on the sign of the diagram at each step. If we define the amount of Monte Carlo steps taken in total as Z_{MC} and the number of steps spent in diagram $\mathcal{A}$ as $Z_{\mathcal{A}}$, then in the statistical limit we see that the probability of visiting diagram $\mathcal{A}$ is equal to

$$\frac{Z_{\mathcal{A}}}{Z_{MC}} = \frac{I_u}{I_T} \quad (4.30)$$

where I_T is defined as the global partition function and is the total absolute weight of the entire configuration space

$$I_T = I_f + \frac{|\lambda|}{\pi} \int |u(|\mathbf{q} - \mathbf{q}'|)u(q')| d\chi dq dq' \quad (4.31)$$

$$+ \frac{|1 + \lambda|}{\pi} \int |u(|\mathbf{q} - \mathbf{q}'|)f(q')| d\chi dq dq' \quad (4.32)$$

$$+ \frac{|\lambda|}{\pi^2} \int |u(|\mathbf{q} - \mathbf{q}'|)u(|\mathbf{q}' - \mathbf{q}''|)f(q'')| d\chi d\chi' dq dq' dq''. \quad (4.33)$$

We then denote the sum of all contributions in a single bin s in the generated histogram as H_s , in the statistical limit it is seen that the distribution $f(q)$ is approximated as

$$\frac{H_s}{Z_{MC}} = \frac{1}{I_T} \int_{q \in \text{bin}_s} f(q) dq \quad (4.34)$$

where bin_s denotes the single bin corresponding to s . In the limit as the bin sizes become infinitesimal we can make the approximation that

$$I_f = \sum_s \left| \int_{q \in \text{bin}_s} f(q) dq \right| \quad (4.35)$$

where this equation is equal to a rearrangement of Equation 4.34 for each histogram bin, summing over all bins results in

$$I_f = \sum_s \left| \int_{q \in \text{bin}_s} f(q) dq \right| = \frac{I_T}{Z_{MC}} \sum_s |H_s|, \quad (4.36)$$

from this equation, we substitute in Equation 4.30 to find an expression for I_f

$$I_f = \frac{\sum_s |H_s|}{Z_{\mathcal{A}}} I_u. \quad (4.37)$$

From here, we are able to produce a value for the scattering length by observing that we can solve Equation 4.14 by evaluating the integral on the right as summations. Furthermore, through the evaluation of the particular function $u(q)$ and the use of Equations 4.34 and 4.30 we find

$$\alpha \rightarrow u(0) + \frac{2I_u}{\pi Z_{\mathcal{A}}} \sum_s u(q_s) H_s. \quad (4.38)$$

This gives us an approximation for the scattering length. We now discuss, in the next section, the results achieved using this method.

4.5 Results and Discussion

An algorithm was set up (in atomic units) for both systems (square well and Yukawa) that used the proposal probabilities stated in Figure 4.2 for each diagram, along with the same acceptance ratios discussed in Section 4.3.

The approach taken to apply BLDMC to both scattering systems was as follows. For a fixed potential value $-V$, the algorithm was run for N Monte Carlo steps, producing a histogram approximation to the scattering amplitude. This histogram was then used for a new simulation at the same potential $-V$, whereby a new histogram was generated from scratch. This procedure was repeated M times and would produce progressively refined histogram approximations for a single potential value. After M approximations, the algorithm would proceed to the next potential value $-V - \Delta V$, where the last histogram generated from the previous potential was carried forward and used as the initial approximation for the new potential. For accurate results it was crucial that histograms were carried forward. This is because the very first histogram approximation uniformly sampled momentum values, providing a crude initial approximation. As a consequence, calculating the scattering length for a particular potential value in regions of divergence required a histogram that depicted that region sufficiently.

The values N (number of steps to generate a histogram) and M (number of histograms generated for a single potential value) were varied to study the impact they had on results. Increasing N , significantly increased computation time, but generally improved accuracy by reducing noise in each histogram approximation. The effect of increasing M was more nuanced, as one would expect increasing M to (generally) always be beneficial. However, for small values of N the likelihood for each histogram to contain more substantial statistical noise was greater. Since each histogram was used self-referentially, a single "bad" histogram could affect all subsequent histograms making errors propagate from one histogram to the next. This was most noticeable in resonance regions, due to the approximations not being able to recover after divergence.

Both systems were implemented with code that is available on GitHub [14]. Furthermore, how values were sampled from $|u(q)|$ and $|f(q)|$ is discussed in Appendix I.

4.5.1 Square Well Results

The reduced Fourier transform of the square well potential is

$$u(q) = \frac{2V_0}{q^3}(\sin q - q \cos q) \quad (4.39)$$

where V_0 can either take negative values (attractive potential) or positive values (repulsive). Furthermore, this equation was used in tandem with the BLDMC method to produce scattering length values for the square well potential.

The convergence parameter λ was varied for the square well potential system to show the effect that different values of λ had on the BLDMC method. This is shown in Figure 4.3, $\lambda = 1$ was omitted from the graph and presented separately in Figure 4.4. For small values of λ , specifically $\lambda < 1$, the method struggled to recover the correct behaviour of the scattering length after the first bound state. Although the approximations converged very well in the beginning and up to the turning point, the method failed completely as the Numerov scattering value diverged to $+\infty$. Unlike $\lambda = 0$, for $\lambda = 0.5$ some approximations did recover after the first bound state, but the resulting values were not accurate. For larger values of λ the calculated values of the scattering length started to “converge” too early, around $V_0 \approx -0.65$ for $\lambda = 2$ and $V_0 \approx -0.875$ for $\lambda = 1.5$. Although, the approximations associated with $\lambda = 1.5$ incorrectly approximated the bound state, it did accurately recover. Overall, Figure 4.3 suggests that the most reliable values of λ , in comparison to the Numerov solutions, lie in the range $0.5 < \lambda < 1.5$. Moreover, to continue approximations after the first bound states, λ must be non-zero.

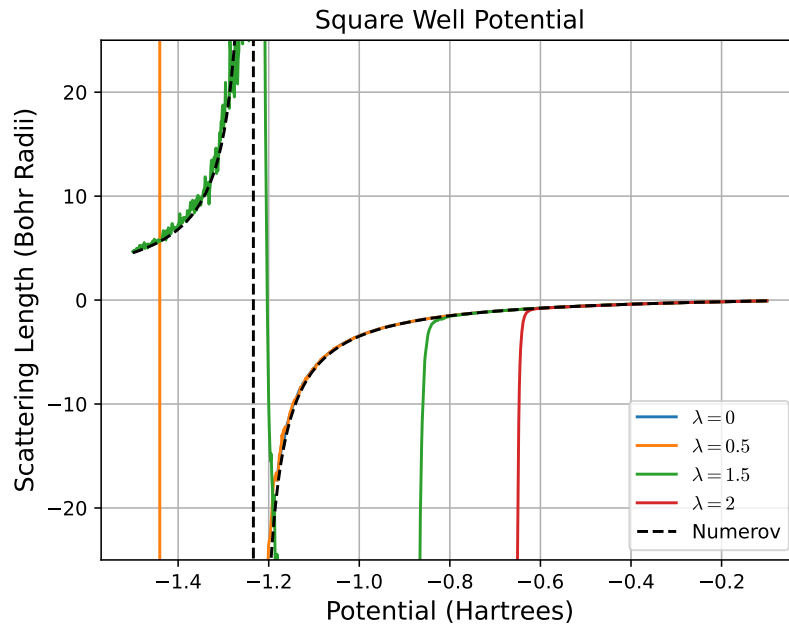


Figure 4.3: The calculation of the scattering length for the square well potential using BLDMC, for varying values of λ . The parameters of the system were: $q_0 = 4$, as the maximum momentum. The number of bins in the generated histograms was 400. The number of steps each simulation had was equal to $N = 1 \times 10^7$, and the number of simulation repeats for a single potential value was $M = 15$. Lastly, the black dashed line shows the scattering length calculated using Numerov’s method.

From Figure 4.4 calculated values show strong convergence for the first bound state, along with the BLDMC method recovering solutions after this bound state. However, for potential values around $V_0 = -3$ it was found that the calculated scattering length diverged and errors grew too

large. To accurately approximate the scattering length with BLDMC beyond potential values of $V_0 = -3$, simulations were started at $V_0 = -2$, making the BLDMC method avoid having to describe the scattering length in the first bound state region. This is shown in Figure 4.5.

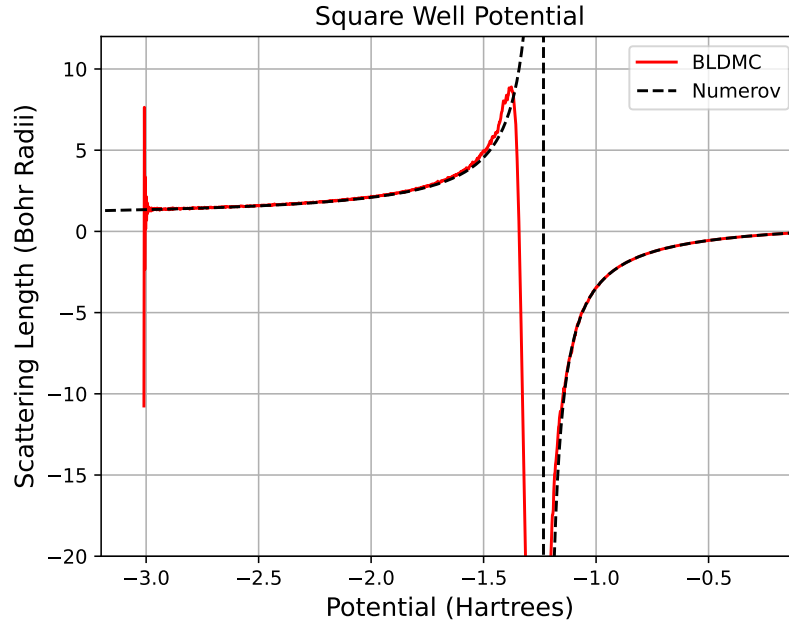


Figure 4.4: The calculation of the scattering length for the square well potential using BLDMC. The parameters of the system were: $N = 1 \times 10^8$ simulations, $M = 15$ number of simulations ran on a single potential, $\lambda = 1$ set as the tuning parameter, $q_0 = 4$ as the maximum momentum value, and the number of bins in the generated histogram was $N_{\text{bins}} = 400$. Lastly, the black dashed line shows the scattering length calculated using Numerov's method.

4.5.2 Yukawa Results

The reduced Fourier transform of the Yukawa potential is:

$$u_y(q) = \frac{2g}{q^2 + 1} \quad (4.40)$$

for $m = 1$ and $\hbar = 1$. The coupling constant g in Equation 4.40 acted as a way to tune potential values. Thus, we show figures in this section that are: scattering length against g .

Figure 4.6 shows that, for a maximum momentum value of $q_0 = 3^1$, the BLDMC method converges accurately to the scattering length obtained from Numerov's method. However, after the first bound state the method does not recover the solution well. Although the calculated scattering length values seem to follow the general trajectory that the Numerov solutions take, they do not converge onto it. By increasing the maximum value of the momentum while

¹The maximum momentum value that the generated histogram could possess.

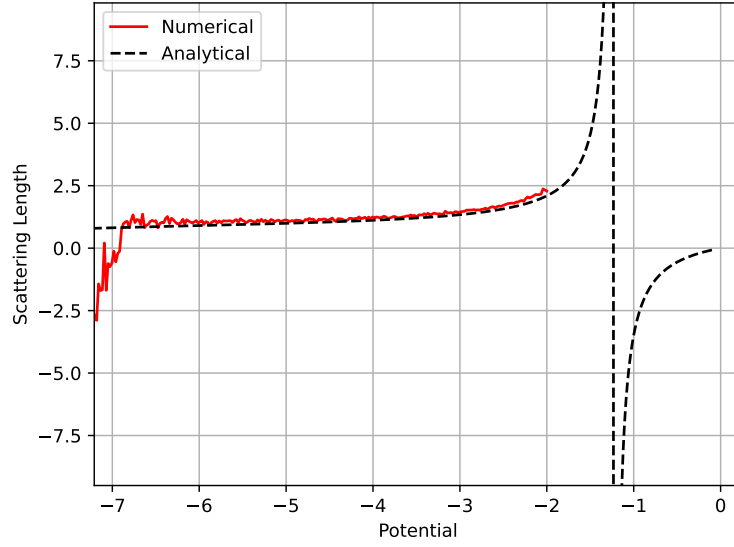


Figure 4.5: The calculation of the scattering length for the square well potential using BLDMC. The potential was given in Hartrees, and the scattering length in Bohr radii. The parameters of the system were: $N = 1 \times 10^7$ simulations, $M = 15$ number of simulations ran on a single potential, $\lambda = 1$ set as the tuning parameter, $q_0 = 3$ as the maximum momentum value, and the number of bins in the generated histogram was $N_{\text{bins}} = 300$. Lastly, the black dashed line shows analytical scattering length (Equation 2.28).

keeping the same number of bins in the approximated histograms, the allowed range of the internal variables was expanded. This is seen in Figure 4.7, whereby the expansion of the allowed momentum values has made convergence better after the first bound state, at the cost of having worse convergence before the bound state.

Finally, convergence for the scattering length using BLDMC was harder for the Yukawa potential as there were multiple bound states closely packed together.

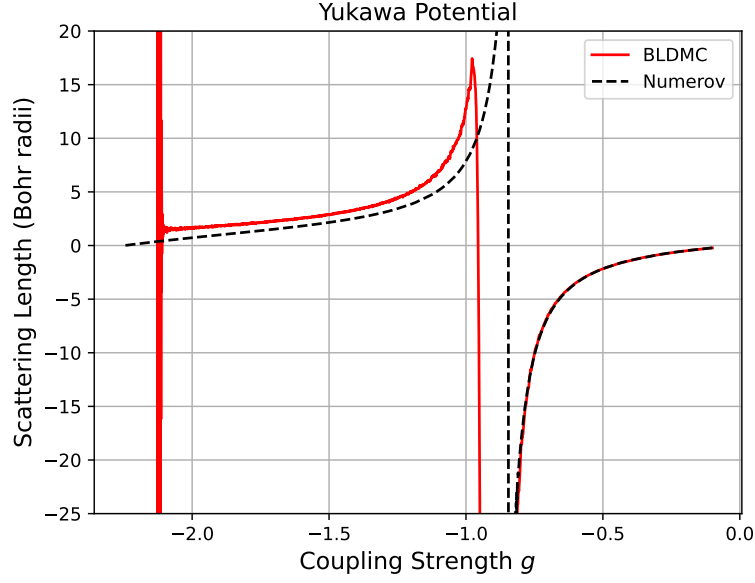


Figure 4.6: The calculation of the scattering length for the Yukawa potential using BLDMC. The parameters of the system were: $N = 1 \times 10^8$ simulations, $M = 15$ number of simulations ran on a single potential, $\lambda = 1$ set as the tuning parameter, $q_0 = 3$ as the maximum momentum value, and the number of bins in the generated histogram was $N_{\text{bins}} = 300$. Lastly, the black dashed line shows the scattering length calculated using Numerov's method.

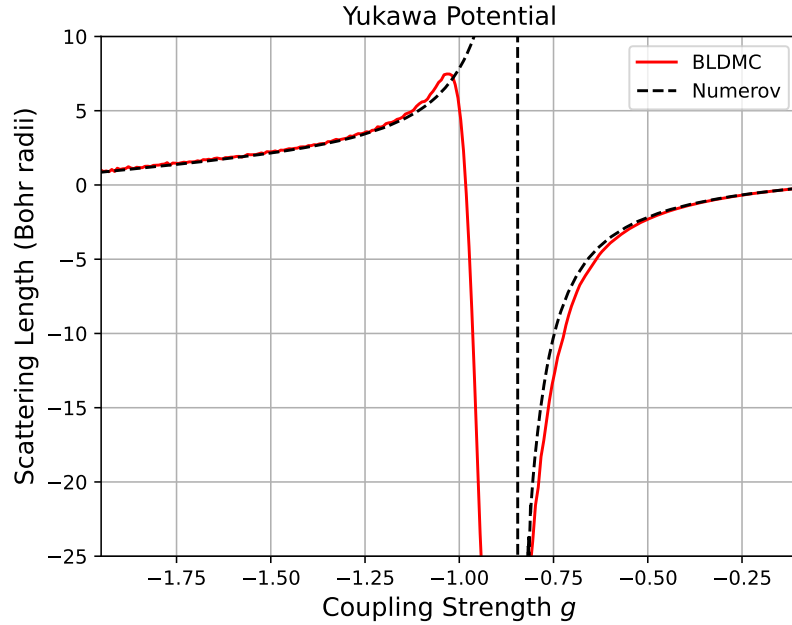


Figure 4.7: The calculation of the scattering length for the Yukawa potential using BLDMC. The parameters of the system were: $N = 1 \times 10^7$ simulations, $M = 10$ number of simulations ran on a single potential, $\lambda = 1$ set as the tuning parameter, $q_0 = 10$ as the maximum momentum value, and the number of bins in the generated histogram was $N_{\text{bins}} = 300$. Lastly, the black dashed line shows the scattering length calculated using Numerov's method.

Chapter 5

Conclusions

We have shown that the bold line diagrammatic Monte Carlo (BLDMC) method can be used to determine functions, such as the scattering amplitude, that are expressed self-consistently in terms of themselves. This is achieved by interpreting each term as a diagram in a configuration space, which was then stochastically sampled from. We applied the BLDMC method to calculate the scattering length for two scattering systems, one governed by the square well potential and the other by the Yukawa potential. These results were then compared with the scattering length values obtained from using Numerov's numerical method. The BLDMC showed strong convergence for both systems up to, including, and surpassing the first bound state. However, for both systems, solutions seemed to diverge shortly after this bound state. In the square well case, the BLDMC method was able to converge scattering length values beyond these diverging points by starting simulations after the first bound state.

We also investigated the λ parameter, introduced in [12], that acted as a tuning tool to help with convergence (Appendix G). It was shown in Chapter 4.5 that without this parameter, the scattering length calculated using BLDMC would not have been able to surpass the first bound state (seen in Figure 4.3). Across both scattering systems, we have demonstrated the potential and usefulness of the diagrammatic Monte Carlo method for evaluating diagrammatic expansions.

Appendix A

Partial Wave and Radial Equation Derivation

We begin by solving for the wavefunction of the Schrödinger equation in spherical coordinates (Equation 2.3). However, first we look at the angular momentum operator $\hat{L}^2$

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2(\theta)} \frac{\partial^2}{\partial \phi^2} \right] \quad (\text{A.1})$$

which has the eigenstate relationship

$$\hat{L}^2 \psi = \hbar^2 l(l+1) \psi, \quad (\text{A.2})$$

this eigenvalue relationship for the angular momentum operator (Equation A.1) remains valid even when the wavefunction is restricted to be independent of ϕ . Using the angular momentum operator (Equation A.1) we can rewrite the Schrödinger equation (Equation 2.3)

$$\left[-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{l(l+1)\hbar^2}{2mr^2} + V(r) \right] \psi(r, \theta) = E \psi(r, \theta). \quad (\text{A.3})$$

Now we assume we can solve Equation A.3 by variable separation: $\psi(r, \theta) = R_l(r) Y_l^0(\theta)$, where $R_l(r)$ is the radial wavefunction and $Y_l^0(\theta)$ are a subset of the spherical harmonics ($m_l = 0$)

$$\left[-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R_l(r)}{\partial r} \right) + \frac{l(l+1)\hbar^2}{2mr^2} R_l(r) + V(r) R_l(r) \right] Y_l^0(\theta) = E R_l(r) Y_l^0(\theta), \quad (\text{A.4})$$

dividing through by $Y_l^0(\theta)$ and separating variables we obtain the radial equation

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R_l(r)}{\partial r} \right) - \frac{l(l+1)}{r^2} R_l(r) \right] + V(r) R_l(r) = E R_l(r), \quad (\text{A.5})$$

which we can further simplify to

$$\left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{l(l+1)}{r^2} - \frac{2mV(r)}{\hbar^2} + k^2 \right) R_l(r) = 0, \quad (\text{A.6})$$

where $k = \sqrt{2mE}/\hbar$. The Y_l^m functions form a complete set on the sphere and can be used to expand any angular function. Therefore, the general solution to the wave function (for $m_l = 0$) is

$$\psi(r, \theta) = \sum_l R_l(r) Y_l^0(\theta) \quad (\text{A.7})$$

where we can replace $Y_l^0(\theta)$ with the Legendre polynomials due to the following relationship

$$Y_l^0(\theta) = \sqrt{\frac{2l+1}{4\pi}} P_l(\cos \theta), \quad (\text{A.8})$$

since the spherical harmonics are proportional to the Legendre polynomials, we can substitute Y_l^0 with $P_l(\cos \theta)$ resulting in

$$\psi(r, \theta) = \sum_l R_l(r) P_l(\cos \theta), \quad (\text{A.9})$$

where the $\sqrt{\frac{2l+1}{4\pi}}$ is absorbed by the radial wavefunction.

Appendix B

Asymptotic Radial Wavefunction and Phase Shift Derivation

We want to simplify the sine and cosine term in Equation 2.11 to a single sine term with a coefficient, which “absorbs” both $B_l(k)$ and $C_l(k)$. We start by expanding $A_l(k) \sin(x + \delta_l)$ where $x = kr - \frac{l\pi}{2}$, while utilising the following relationship

$$\sin(A + B) = \sin(A) \cos(B) + \sin(B) \cos(A). \quad (\text{B.1})$$

Expanding $A_l(k) \sin(x + \delta_l)$ yields

$$A_l(k) \sin(x + \delta_l) = A_l(\sin x \cos \delta_l + \sin \delta_l \cos x) \quad (\text{B.2})$$

comparing with the expression inside the square brackets in Equation 2.11, Equation B.2 has the same form and both equations can be set equal to each other. Comparing coefficients results in

$$B_l(k) = A_l(k) \cos(\delta_l) \quad (\text{B.3})$$

$$-C_l(k) = A_l(k) \sin(\delta_l), \quad (\text{B.4})$$

we solve for $A_l(k)$ by squaring each Equation B.3 & B.4 and adding them

$$B_l^2(k) + (-C_l(k))^2 = (A_l(k) \cos \delta_l)^2 + (A_l(k) \sin \delta_l)^2 \quad (\text{B.5})$$

$$A_l(k) = \sqrt{B_l^2(k) + C_l^2(k)}. \quad (\text{B.6})$$

Dividing Equation B.4 by Equation B.3 leads to an expression for the phase shift

$$-\frac{C_l(k)}{B_l(k)} = \tan(\delta_l) \implies \delta_l(k) = \arctan\left(-\frac{C_l(k)}{B_l(k)}\right), \quad (\text{B.7})$$

since $\arctan(x)$ is an odd function, $\delta_l(k)$ can be expressed as

$$\delta_l(k) = -\arctan\left(\frac{C_l(k)}{B_l(k)}\right). \quad (\text{B.8})$$

Appendix C

Square Well Scattering Length Derivation

Starting from Equation 2.20 we set $l = 0$ and replace $x = ka$ and $y = Ka$ for simplicity. We then substitute in the spherical Bessel and Neumann functions and their derivatives for $l = 0$ (Equations 2.22, 2.23, 2.24, and 2.25) to get

$$\tan \delta_0(k) = \frac{k \left(\frac{x \cos(x) - \sin(x)}{x^2} \right) \left(\frac{\sin y}{y} \right) - K \left(\frac{\sin x}{x} \right) \left(\frac{y \cos(y) - \sin(y)}{y^2} \right)}{k \left(\frac{x \sin(x) + \cos(x)}{x^2} \right) \left(\frac{\sin y}{y} \right) + K \left(\frac{\cos x}{x} \right) \left(\frac{y \cos(y) - \sin(y)}{y^2} \right)}, \quad (\text{C.1})$$

we then multiple the numerator and denominator by $xy/(\cos x \cos y)$

$$= \frac{ky \left(x - \tan(x) \right) \left(\tan y \right) - Kx \left(\tan x \right) \left(y - \tan(y) \right)}{ky \left(x \tan(x) + 1 \right) \left(\tan y \right) + Kx \left(y - \tan(y) \right)}, \quad (\text{C.2})$$

substituting back in $ka \equiv x$ and $Ka \equiv y$ allows for mixed terms to cancel

$$= \frac{kKa \left(ka - \tan(ka) \right) \left(\tan(Ka) \right) - Kka \left(\tan(ka) \right) \left(Ka - \tan(Ka) \right)}{kKa \left(ka \tan(ka) + 1 \right) \left(\tan(Ka) \right) + Kka \left(Ka - \tan(Ka) \right)} \quad (\text{C.3})$$

$$= \frac{kKa^2 \left(k \tan(Ka) - K \tan(ka) \right)}{kKa^2 \left(k \tan(ka) \tan(Ka) + K \right)}. \quad (\text{C.4})$$

Finally, we have a simplified form of $\tan \delta_0(k)$ for $l = 0$

$$\tan \delta_0(k) = \frac{k \tan(Ka) - K \tan(ka)}{K + k \tan(ka) \tan(Ka)}. \quad (\text{C.5})$$

From here, we work with small values of k , i.e. near $k = 0$. Multiplying Equation C.5 by k and taking the reciprocal we get

$$k \cot \delta_0(k) = k \frac{K + k \tan(ka) \tan(Ka)}{k \tan(Ka) - K \tan(ka)}, \quad (\text{C.6})$$

we expand Equation C.6 using the small angle approximation ($\tan(ka) \approx ka$) and relabel $Ka = \gamma$ when $k = 0$

$$k \cot \delta_0(k) = \frac{kK + k^3 a \tan(\gamma)}{k(\tan(\gamma) - \gamma)}, \quad (\text{C.7})$$

we remove the k^3 term as this is considered negligible and K is replaced with $K = \gamma/a$

$$k \cot \delta_0(k) = \frac{1}{a} \frac{\gamma}{\tan(\gamma) - \gamma}. \quad (\text{C.8})$$

The scattering length definition is [6]

$$\alpha = - \lim_{k \rightarrow 0} \frac{\tan \delta_0(k)}{k} \quad (\text{C.9})$$

using this definition of the scattering length along with Equation C.8 we find

$$k \cot \delta_0(k) = -\frac{1}{\alpha} + \frac{1}{2} r_0 k^2 + \dots \quad (\text{C.10})$$

where r_0 is the effective range. Substituting Equation C.8 into Equation C.10 produces the analytical scattering length

$$\alpha = \left(1 - \frac{\tan \gamma}{\gamma}\right) a. \quad (\text{C.11})$$

Appendix D

Numerov's Method Derivation

Here, we derive Numerov's method for a general second-order differential equation with no first-order terms

$$\frac{d^2 y}{dx^2} + g(x)y = s(x), \quad (\text{D.1})$$

we take the Taylor expansion of $y(x)$ about the point x_0

$$\begin{aligned} y(x) = & y(x_0) + (x - x_0)y^{(1)}(x_0) + \frac{(x - x_0)^2}{2!}y^{(2)}(x_0) + \frac{(x - x_0)^3}{3!}y^{(3)}(x_0) \\ & + \frac{(x - x_0)^4}{4!}y^{(4)}(x_0) + \frac{(x - x_0)^5}{5!}y^{(5)}(x_0) + \mathcal{O}((x - x_0)^6) \end{aligned} \quad (\text{D.2})$$

where $y^{(i)}$ is the i th derivative with respect to x and $\mathcal{O}((x - x_0)^6)$ shows the truncation error. We denote the distance $x - x_0$ by $h = x - x_0$, and substitute it into Equation D.2 to get

$$y(x_0 + h) = y(x_0) + hy^{(1)}(x_0) + \frac{h^2}{2!}y^{(2)}(x_0) + \frac{h^3}{3!}y^{(3)}(x_0) + \frac{h^4}{4!}y^{(4)}(x_0) + \frac{h^5}{5!}y^{(5)}(x_0) + \mathcal{O}(h^6). \quad (\text{D.3})$$

We discretise space into uniform points, evenly spaced by h . By changing $h = x_{n+1} - x_n$, we obtain an expression that relates y_n and y_{n+1}

$$y_{n+1} = y_n + hy_n^{(1)} + \frac{h^2}{2!}y_n^{(2)} + \frac{h^3}{3!}y_n^{(3)} + \frac{h^4}{4!}y_n^{(4)} + \frac{h^5}{5!}y_n^{(5)} + \mathcal{O}(h^6). \quad (\text{D.4})$$

where $y_n \equiv y(x_n)$. Replacing h with $h = x_n - x_{n-1}$ we get a backward step rather than a step forward in our discretised space. However, $h = x_n - x_{n-1}$ is effectively the same as multiplying $h = x_{n+1} - x_n$ from Equation D.4 by -1 . Therefore, we get the following result

$$y_{n-1} = y_n - hy_n^{(1)} + \frac{h^2}{2!}y_n^{(2)} - \frac{h^3}{3!}y_n^{(3)} + \frac{h^4}{4!}y_n^{(4)} - \frac{h^5}{5!}y_n^{(5)} + \mathcal{O}(h^6) \quad (\text{D.5})$$

By summing Equations D.4 & D.5 we cancel out odd powers

$$y_{n+1} - 2y_n + y_{n-1} = h^2 y_n^{(2)} + \frac{h^4}{12} y_n^{(4)} + \mathcal{O}(h^6). \quad (\text{D.6})$$

To get expression for $y_n^{(2)}$ and $y_n^{(4)}$ we use the general differential equation we started with (Equation D.1) for $y_n^{(2)}$, and approximate it using the same analysis as above for $y_n^{(4)}$. Doing this results in

$$y_n^{(4)} = \frac{d^2}{dx^2}(-g_n y_n + s_n) = \frac{-g_{n+1} y_{n+1} + s_{n+1} + 2g_n y_n - 2s_n - g_{n-1} y_{n-1} + s_{n-1}}{h^2} + \mathcal{O}(h^4) \quad (\text{D.7})$$

Substituting Equation D.7 and $y_n^{(2)} = -g_n y_n + s_n$ into Equation D.6 results in

$$\left(1 + \frac{h^2}{12} g_{n+1}\right) y_{n+1} - 2\left(1 - \frac{5h^2}{12} g_n\right) y_n + \left(1 + \frac{h^2}{12} g_{n-1}\right) y_{n-1} \quad (\text{D.8})$$

$$= \frac{h^2}{12} (s_{n+1} + 10s_n + s_{n-1}) + \mathcal{O}(h^6) \quad (\text{D.9})$$

Relating this to the reduced radial equation (2.29), we see that $s(x) = 0$, the expression included in square brackets corresponds to $g(x)$ from Equation D.1, and y is u_l . Rearranging for y_{n+1} , we obtain

$$y_{n+1} = \frac{2\left(1 - \frac{5h^2}{12} g_n\right) y_n - \left(1 + \frac{h^2}{12} g_{n-1}\right) y_{n-1}}{\left(1 + \frac{h^2}{12} g_{n+1}\right)} + \mathcal{O}(h^6). \quad (\text{D.10})$$

D.1 Numerov Initialisation

We initialise the first two solutions u_0 and u_1 by examining the reduced radial equation (2.29) at small values of r . In this limit, the centrifugal term dominates, resulting in the differential equation simplifying to

$$\frac{d^2 u_l}{dr^2} = \frac{l(l+1)}{r^2} u_l(r). \quad (\text{D.11})$$

We solve this differential equation by assuming $u_l(r) = r^\lambda$. Substituting yields

$$\lambda(\lambda-1)r^{\lambda-2} = l(l+1)r^{\lambda-2}, \quad (\text{D.12})$$

$r^{\lambda-2}$ cancels on either side, and we solve for λ

$$\lambda^2 - \lambda - l(l+1) = 0 \quad (\text{D.13})$$

$$\lambda_1 = l+1 \quad \text{or} \quad \lambda_2 = -l. \quad (\text{D.14})$$

Solution λ_2 is discarded due to it diverging as $r \rightarrow 0$, leading to a non-physical wavefunction: $u_l = 1/r^l \rightarrow \infty$ as $r \rightarrow 0$. Therefore, we choose λ_1 , this gives a solution

$$u_l(r) \sim r^{l+1}. \quad (\text{D.15})$$

Evaluating Equation D.15 at the first grid point $r = h$, the first two points are initialised as

$$u_0 = 0 \quad \text{and} \quad u_1 = h^{l+1}. \quad (\text{D.16})$$

D.2 Numerov Pseudo Code

```

1 Initialise radial grid:
2   r_values = [0, h, 2h, ..., r_max]
3   n = length(r_values)
4
5 Initialise solution array:
6   u = [0, 0, ..., 0] # same length as r_values
7
8 Set boundary conditions:
9   u[0] = 0
10  u[1] = h^(l+1)
11
12 Define:
13   A = h^2 / 12
14
15 For i = 1 to n-2:
16
17   Compute:
18     g_{i-1} = g(r_values[i-1])
19     g_i     = g(r_values[i])
20     g_{i+1} = g(r_values[i+1])
21
22   where:
23     g(r) = (2mE) - l(l+1)/r^2 - 2mV(r) # hbar = 1
24
25   Update:
26     u[i+1] =
27       (2(1 - 5Ag_i) u[i] - (1 + Ag_{i-1}) u[i-1]) / (1 + Ag_{i+1})
28
29 Output:
30   u, r_values

```

Appendix E

Numerical Phase shift Derivation

We begin by dividing Equation 2.32 by Equation 2.31

$$\frac{u_l(r_q)}{u_l(r_p)} = \frac{r_q [\cos(\delta_l)j_l(kr_q) - \sin(\delta_l)n_l(kr_q)]}{r_p [\cos(\delta_l)j_l(kr_p) - \sin(\delta_l)n_l(kr_p)]} \quad (\text{E.1})$$

from this we define the ratio

$$\frac{r_p u_l(r_q)}{r_q u_l(r_p)} = G = \frac{[\cos(\delta_l)j_l(kr_q) - \sin(\delta_l)n_l(kr_q)]}{[\cos(\delta_l)j_l(kr_p) - \sin(\delta_l)n_l(kr_p)]} \quad (\text{E.2})$$

multiplying by the denominator and grouping terms we get

$$\cos(\delta_l)[Gj_l(kr_p) - j_l(kr_q)] = \sin(\delta_l)[Gn_l(kr_p) - n_l(kr_q)] \quad (\text{E.3})$$

dividing by $\cos(\delta_l)$ and rearranging gives

$$\tan(\delta_l) = \frac{Gj_l(kr_p) - j_l(kr_q)}{Gn_l(kr_p) - n_l(kr_q)} \quad (\text{E.4})$$

and

$$\delta_l = \arctan \left(\frac{Gj_l(kr_p) - j_l(kr_q)}{Gn_l(kr_p) - n_l(kr_q)} \right). \quad (\text{E.5})$$

Appendix F

Metropolis-Hastings DiagMC Trick Detailed Balance Equation Derivation

We begin with the general detailed balance equation for the system described in Section 3.3 using the weights from Equations 3.40 and 3.41

$$f(x_a)dx \cdot P_{1 \rightarrow 0}(x_a) = D_0 \cdot dP_{0 \rightarrow 1}(x_a). \quad (\text{F.1})$$

We now use how the system handles transfer updates from Type-0 to Type-1 and vice versa, this is described Chapter 3.3, the probabilities are again broken down into proposal and acceptance components (seen in Chapter 3.2.1). From this we obtain the following equations

$$dP_{0 \rightarrow 1}(x_a) = p_{0 \rightarrow 1}^{(\text{propose})} \cdot \Omega(x_a)dx \cdot P_{0 \rightarrow 1}^{(\text{accept})}(x_a) \quad (\text{F.2})$$

$$P_{1 \rightarrow 0}(x_a) = p_{1 \rightarrow 0}^{(\text{propose})} P_{1 \rightarrow 0}^{(\text{accept})}(x_a) \quad (\text{F.3})$$

substituting these Equations F.2 and F.3 into the detailed balance equation (F.1) results in

$$f(x_a)p_{1 \rightarrow 0}^{(\text{propose})}P_{1 \rightarrow 0}^{(\text{accept})}(x_a) = D_0p_{0 \rightarrow 1}^{(\text{propose})}\Omega(x_a)P_{0 \rightarrow 1}^{(\text{accept})}(x_a). \quad (\text{F.4})$$

Appendix G

Lambda Tuning Parameter

The parameter λ acts as a tuning parameter that does not change the exact solution of the original equation, but rather complements convergence by changing the effective operator, such that the error propagation is no longer governed solely by the eigenvalues a_i of $\mathbf{A}$. To understand its impact we start from Equation 4.20, where we expand the equation in terms of the eigenbasis of $\mathbf{A}$ with eigenvectors $\{|\phi_i\rangle\}$, where we assume $\mathbf{A}$ is Hermitian. We also express f as an iterative process, such that $f^{(n)}$ produces the next term $f^{(n+1)}$

$$f_i^{(n+1)} = b_i + (1 + \lambda)a_i f_i^{(n)} - \lambda a_i b_i - \lambda a_i^2 f_i^{(n)}. \quad (\text{G.1})$$

We then define the error in each approximation n as $\delta_i^{(n)} = f_i - f_i^{(n)}$ where f_i is the solution. Rewriting Equation G.1 in terms of this error yields

$$\delta_i^{(n+1)} = [(1 + \lambda)a_i - \lambda a_i^2] \delta_i^{(n)} \quad (\text{G.2})$$

We see that for errors to reduce we require that the coefficient $|(1 + \lambda)a_i - \lambda a_i^2| < 1$. Furthermore, we can express error as the cumulative error

$$\delta_i^{(n)} = \delta_i^{(0)} [(1 + \lambda)a_i - \lambda a_i^2]^n, \quad (\text{G.3})$$

and as $n \rightarrow \infty$ the iterative approximation converges to the solution f_i , provided $|(1 + \lambda)a_i - \lambda a_i^2| < 1$ holds true. For $\lambda = 0$ the error takes the form

$$\delta_i^{(n+1)} = [a_i] \delta_i^{(n)} \quad (\text{G.4})$$

where we see that the errors depend only on the eigenvalues of $\mathbf{A}$ for $\lambda = 0$.

With the addition of this λ parameter we have shown that in areas where naive iteration (Equation 4.15) may have failed to converge, the λ parameter offers an alternative convergence tool.

Appendix H

BLDMC: Acceptance Ratio Derivation

Here, we derive the acceptance ratio $R_{\mathcal{A} \rightarrow \mathcal{C}}$, where the same logic follows for the rest of the acceptance ratios. We start by defining the weights W of each diagram

$$W_{\mathcal{A}}(q) = |u(q)| \quad (\text{H.1})$$

$$W_{\mathcal{C}}(q, q', \chi) = \frac{|\lambda|}{\pi} |u(|\mathbf{q} - \mathbf{q}'|)u(q')| \quad (\text{H.2})$$

using Equation 3.29, the move from diagram $\mathcal{A} \rightarrow \mathcal{C}$ has the proposal split into three pieces:

1. Proposal to swap: $p_{\mathcal{AC}}$
2. Sample χ uniformly on the interval $[-1, 1]$ resulting in a density of $\frac{1}{2}$
3. Sampling q from $|u(q')|/I_u$, where I_u is defined in Equation 4.22

This results in $\Omega(\mathcal{A}, \mathcal{C})$ being equal to

$$\Omega(\mathcal{A}, \mathcal{C}) = p_{\mathcal{AC}} \cdot \frac{1}{2} \cdot \frac{|u(q')|}{I_u}. \quad (\text{H.3})$$

the equivalent for the move $\mathcal{C} \rightarrow \mathcal{A}$ is

$$\Omega(\mathcal{C}, \mathcal{A}) = p_{\mathcal{CA}} \quad (\text{H.4})$$

Therefore, substituting $\Omega(\mathcal{A}, \mathcal{C})$, $\Omega(\mathcal{C}, \mathcal{A})$, $W_{\mathcal{A}}$, and $W_{\mathcal{C}}$ from Equations H.3, H.4, H.1, and H.2 respectively, into

$$R_{\mathcal{A} \rightarrow \mathcal{C}} = \frac{W_{\mathcal{C}}\Omega(\mathcal{C}, \mathcal{A})}{W_{\mathcal{A}}\Omega(\mathcal{A}, \mathcal{C})} \quad (\text{H.5})$$

yields

$$R_{\mathcal{A} \rightarrow \mathcal{C}} = \frac{2|\lambda|I_u p_{\mathcal{CA}}}{\pi p_{\mathcal{AC}}} \left| \frac{u(|\mathbf{q} - \mathbf{q}'|)}{u(q)} \right|. \quad (\text{H.6})$$

From this procedure all other acceptance ratios can be obtained. These ratios are also seen from the paper [12] where Prokof'ev et al. have set up the same scattering system. However, they have mislabeled diagrams.

Appendix I

BLDMC: Sampling Values from Distributions

In practice, how we generated samples from the distribution: $|u(q)|$, was by building weights associated with $|u(q)|$. This involved making an array, that we denote W_u . The length of this array was the same size as the number of bins N_{bins} in each approximation histogram. The array would look similar to the following

$$W = [|u(q_1)|\Delta q, |u(q_2)|\Delta q, \dots, |u(q_{N_{\text{bins}}})|\Delta q]. \quad (\text{I.1})$$

where Δq is equivalent to $\Delta q = |q_s - q_{s-1}|$ where s is the index related to the bin. The weights in Equation I.1 were then added sequentially to create a cumulative distribution function (CDF) denoted C_u . If we denote w_i as the i th weight in Equation I.1, where $w_i = |u(q_i)|\Delta q$, C_u can be expressed as

$$C_u = [w_1, w_1 + w_2, w_1 + w_2 + w_3, \dots]. \quad (\text{I.2})$$

Then the total (T_u) is the sum of all the weights

$$T_u = \sum_{i=1}^{N_{\text{bins}}} w_i. \quad (\text{I.3})$$

A random element would be selected from C_u (Equation I.2) using rT_u where $r \in [0, 1]$, we denote this random element as X . The index of this random value X was then found by performing a method similar to bisection root finding. The index was then multiplied by Δq to find the q value that was distributed from $|u(q)|$. A similar structure was implemented for $|f(q)|$, where the weights were instead the absolute value of each bin.

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